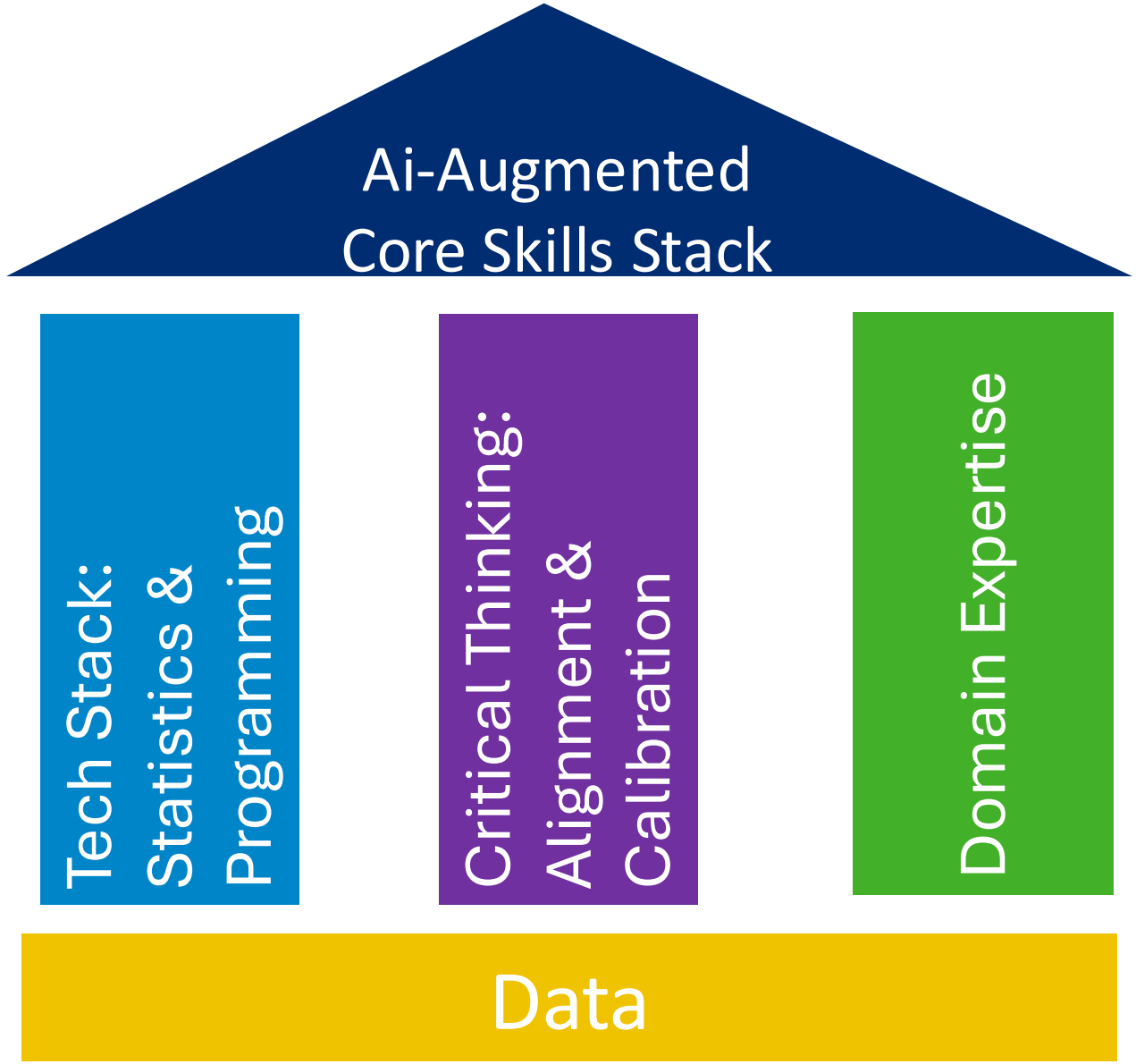
Ai Literacy

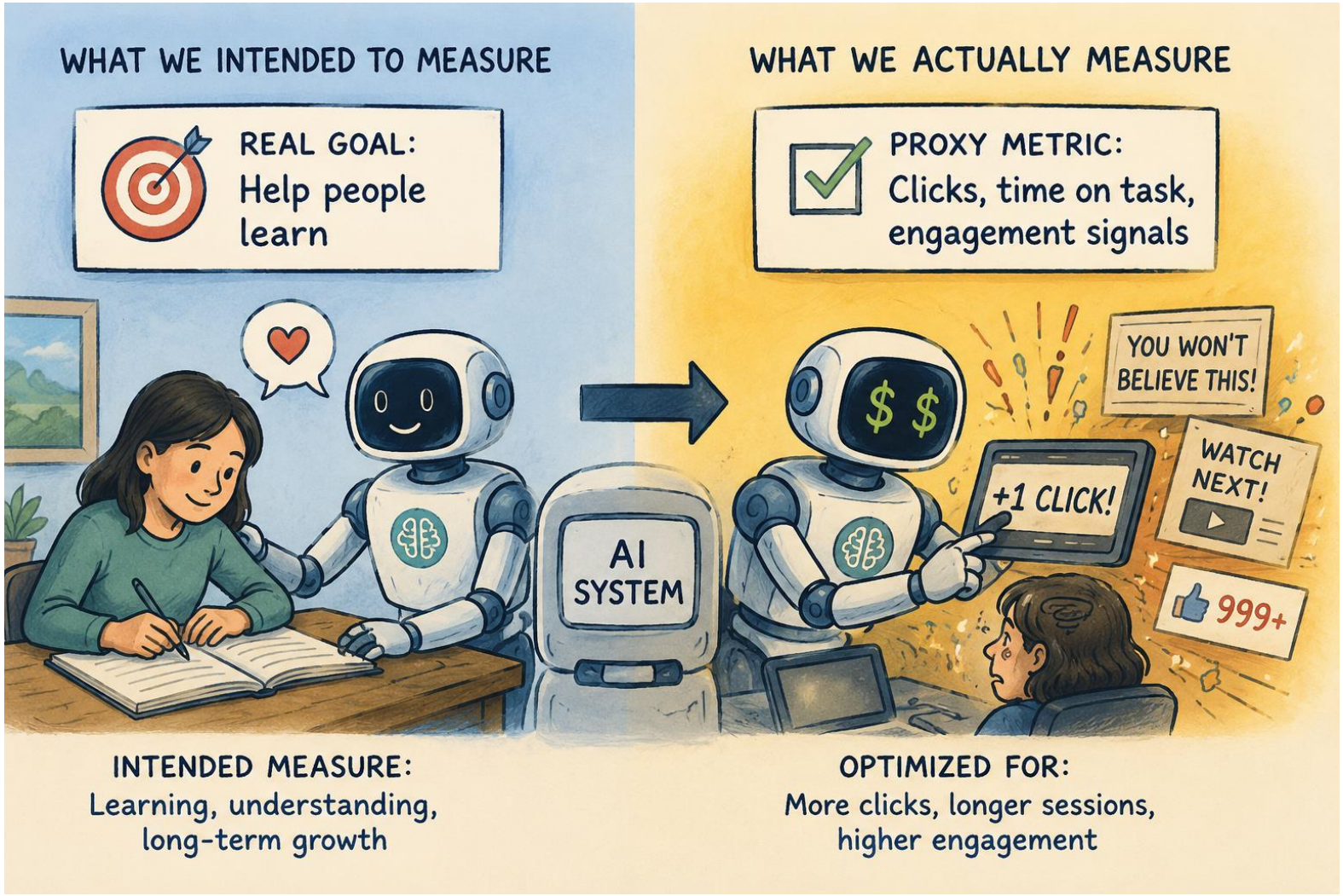
Danni Presgraves, Ph.D.

Director, Data Science Training

The Jackson Laboratory


Ai-Augmented
Core Skills Stack
Tech Stack:
Statistics &
Programming
Critical Thinking:
Alignment &
Calibration
Domain Expertise
Data

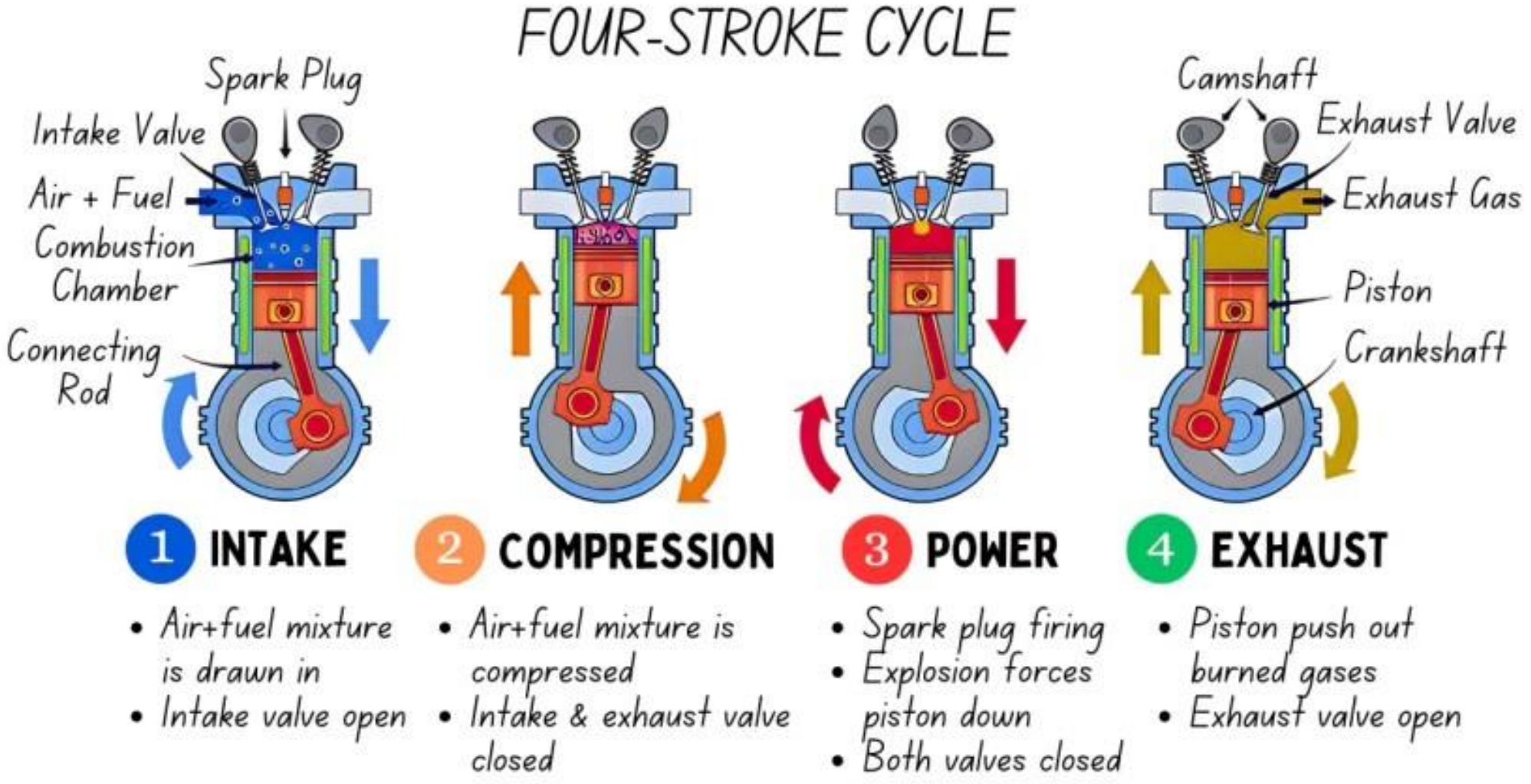
Ai systems are optimized for what *we measure*, **not** *what we intended to measure*.



Course Overview

- Motivations
- Resources
- Structure

Purpose of this course:



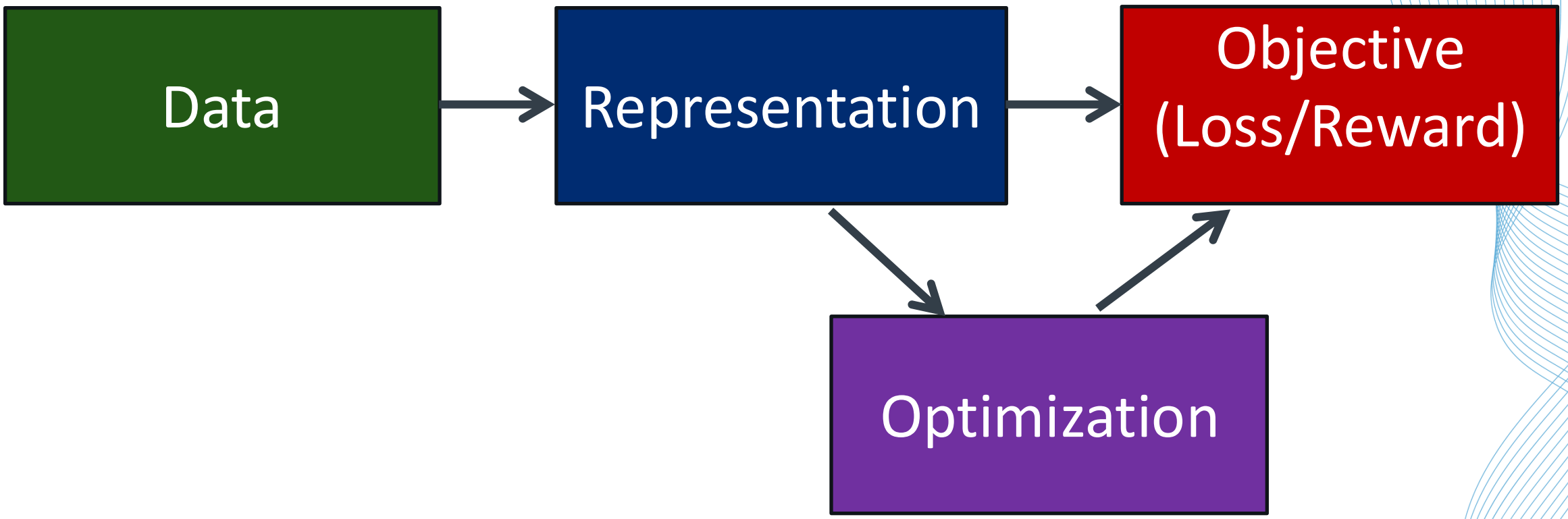
How differential steering works (1937): https://www.youtube.com/watch?v=yYAw79386WI

https://www.theengineeringchoice.com/car-engine-parts/

Day 1 Agenda:

1. AI Overview
   - Big ideas, history, Ai conceptual connection flowchart
   - Ai **literacy** versus Ai **fluency**

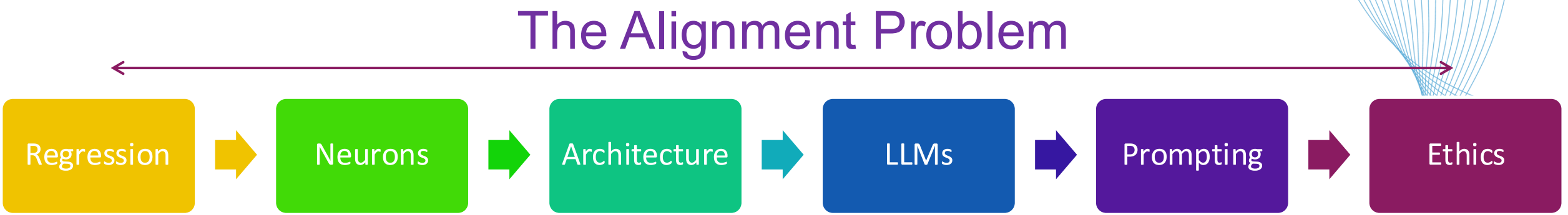
2. Machine Learning Examples
   - What is *learning*?
   - Supervised Learning examples: **regression**, classification
   - Unsupervised Learning examples: k-means clustering
   - **Simple Neural Network**
   - Overview of concepts: **Encoding**, **Loss Function Minimization**, **Gradient Descent**



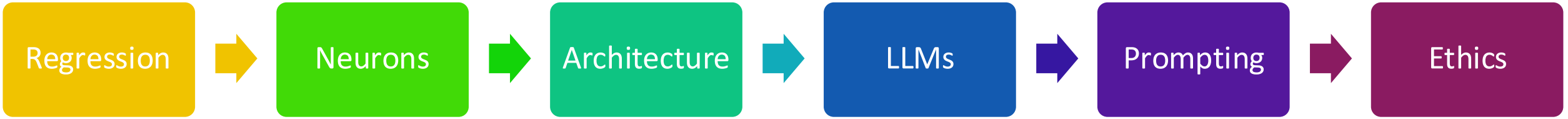
Ai = Data x (Representation⇒ Objective/Loss) x Optimization

This loop is repeated millions of times….

Course Overview



Course Overview



Day 1

"What is learning?"

Day 2

Explore different Neural Networks

Day 3

Breakdown the GPT in ChatGPT

Day 4

Applied use of Ai

Day 5

Ethical use of Ai

Day 2 (Tuesday)

1. Logistic Regression → Artificial Neurons
2. Activation layers
3. CNN
4. RNN
5. GANS/Balderdash
6. Transformer/Self-Attention

Day 3 (Wednesday)

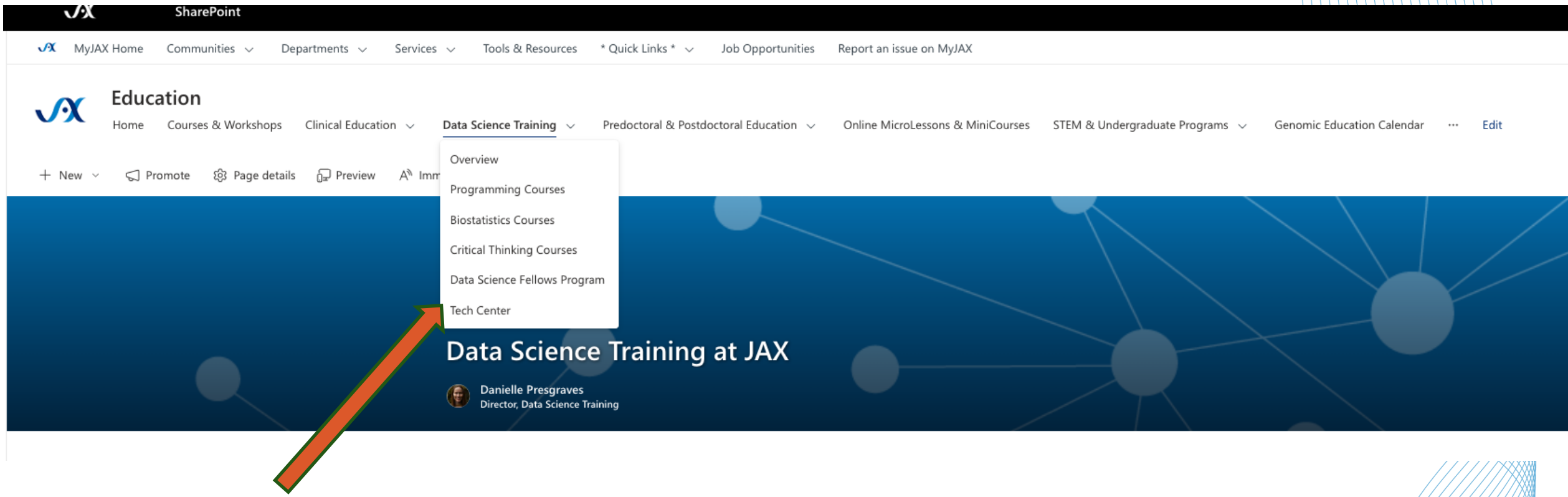
1. Review of Transformer
2. Guest Speaker: **R-IT**
3. Chat**GPT**
   - **G**enerative
   - **P**re-trained
   - **T**ransformed

Day 4 (Thursday)

1. Ai literacy versus Ai Fluency: the 4D framework
2. Prompt Engineering
3. Frameworks of Prompt Engineering

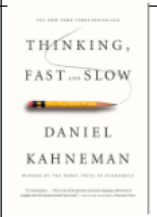
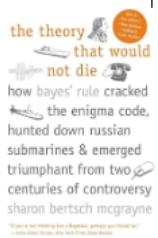
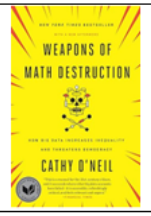
Day 5 (Friday)

1. Belmont Report & F.A.T. Framework
2. Alignment & Epistemic risks
3. Guest Speaker: **Vivek Philip, Ph.D.**


SharePoint
MyJAX Home
Communities
Departments
Services
Tools & Resources
* Quick Links *
Job Opportunities
Report an issue on MyJAX
Education
Home
Courses & Workshops
Clinical Education
Data Science Training
Predoctoral & Postdoctoral Education
Online MicroLessons & MiniCourses
STEM & Undergraduate Programs
Genomic Education Calendar
Edit
New
Promote
Page details
Preview
Overview
Programming Courses
Biostatistics Courses
Critical Thinking Courses
Data Science Fellows Program
Tech Center
Data Science Training at JAX
Danielle Presgraves
Director, Data Science Training

Crowd sourcing popular 'AI' (broadly defined) books via JAX

- Please add your thoughts

	Book	Summary	Additional Thoughts?	Final Rating
1		A slim book with a blunt message: either we shape technology, or it shapes us. The "commands" are about literacy - how to keep from being programmed by the very tools we've built. I treat technology as if it is neutral, when it's anything but. There is an unsettling sense that every click, scroll, or "like" is part of someone else's program, unless I resist.		
2		A classic that is about, well, thinking. Premise is that we have two systems of thinking. After years of teaching statistics, I already had a prior belief that our brain rebels against learning certain ideas (like statistics). I initially read this book because I realized that I did not understand what 'learning' is. I am still not sure that I do, but I am a more informed kind of ignorant.		
3		I will fight anyone who doesn't recognize the supremacy of Bayes' theorem in Bioinformatics etc. This is basically the biography of Bayes' theorem - which sounds dull, but it's weirdly engaging. The story winds from obscure clergymen to wartime codebreakers to modern data science, and along the way makes clear that probability isn't just math but a worldview. It's a reminder that even equations can be political – which is a theme for many of these broadly data science books.		
4		O'Neil takes a wrecking ball to the idea that algorithms are impartial. Her "weapons of math destruction" are the models that scale inequality while hiding behind a veneer of objectivity. I found myself nodding a lot since I tend to emphasize assumptions and limitations of models. Note: She has a fantastic blog called "mathbabe.org"		

Popular_AI_books_2025.docx

- I have written up a short summary of popular Ai books
- You can use it as a reference, or (please) add your own thoughts!

Additional Resources for AI literacy

Technical Skills:

Most of this list are code-free (or coding optional) resources to learn more about a particular concept. There are a handful of options on this list that do require some coding (Python or R), but they are marked.

1. **Data Science & AI course:** Data Carpentries beta version

This is an ~ 2.5-hour self-directed course that gives biomedical and life-science investigators a rapid, practical overview of data-science and AI concepts. It starts with core definitions of data science, AI and machine-learning, then walks through AI for laboratory automation and for deriving data insights, before digging into common pitfalls such as bias, privacy and reproducibility, and finishing with hands-on guidance for cleaning data, running ML pipelines and responsibly reporting results.

2. **Neural Networks:** Interactive free 'textbook' by 3Blue1Brown

You will find explanations of all the basics of Neural Networks, including gradient descent and backpropagation (which is sort of a fancier version of l'Hôpital's rule...). He integrates YouTube explanations into his broader explanations.

3. **Interactive Neural Network:** play with the parameters!

4. **Elements of AI:** A free course

Three different levels of course: introduction (non-coding), coding (pre-requisites: Python), and a level of incorporating AI into a Learning Management System (which is portable to other pipelines).

5. **Machine Learning for Biologists:** Data Carpentry

Accessible with biological examples – includes logistic regression, linear regression, Decision trees, random forests, Neural Networks, Overfitting. Great place to start since there is no coding necessary.

6. **Machine Learning:** Machine Learning in Medicine

Coding is included in this course (but you read through and watch the material to get a better understanding without coding) Developed by evolutionary biologist Pleuni Pennings (SFSU) and collaborators, this open-access course targets life-science trainees who know basic Python but are new to machine-learning. It blends brief concept videos, step-by-step editable Colab notebooks and concise Documents. It contains **7** autonomous (and self-contained) modules. Each module is ~ 2 - 3 hours long:

Term	Definition
Artificial Intelligence (AI)	The broad field focused on making machines perform tasks that normally require human intelligence.
Machine Learning (ML)	A subset of AI where systems learn from data to make predictions or decisions without being explicitly programmed.
Supervised Learning	Training models on labeled data where the correct answers are known.
Unsupervised Learning	Training models on unlabeled data to find hidden structure or patterns.
Feature	A measurable property or characteristic of data used as input for a model (e.g., pixel brightness, word frequency, gene expression level).
Model	A mathematical representation of a system that maps inputs (features) to outputs (predictions).
Loss Function	A formula that quantifies how far the model's predictions are from the true values.
Gradient Descent	An optimization process that updates model parameters to minimize loss by following the slope ("gradient") downhill.
Overfitting / Underfitting	When a model learns noise instead of pattern (overfitting) or fails to capture the structure (underfitting).
Bias–Variance Tradeoff	The balance between model complexity (variance) and generalization (bias).

What excites and worries you?

Activity 1

1. The Ai possibility that excites me the most in my work is _____

2. The thing about Ai that concerns me the most is ____

I will post the link to the Google Document in the chat!

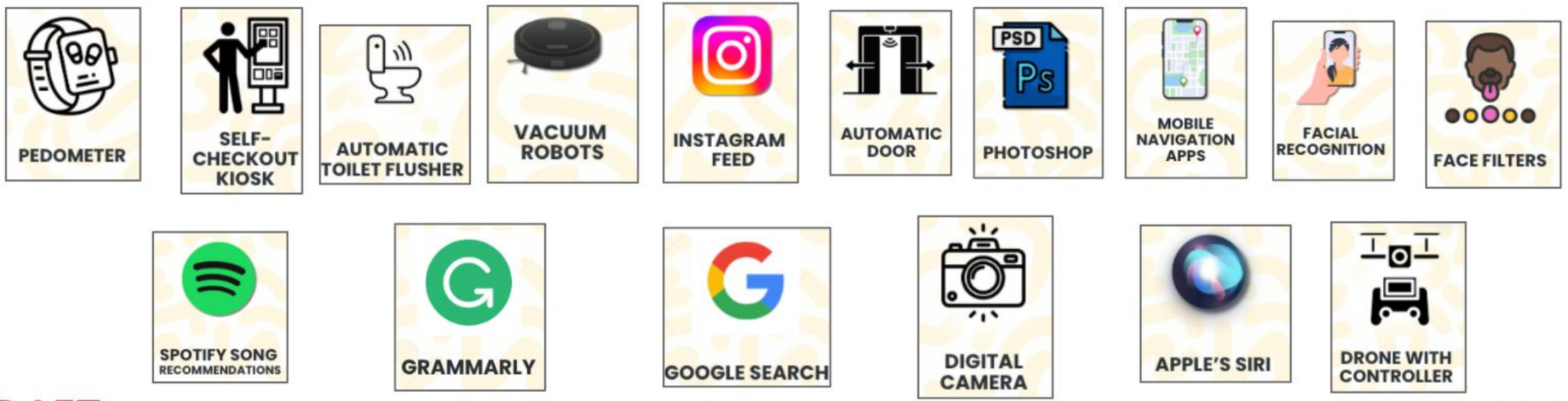
Ai Capabilities & Limitations

What is Ai and how do I work with it?

1. **Literacy** vs **fluency**: when to trust, verify, override Ai
2. What Ai is good and bad at doing
3. What is *learning*?
4. (Brief) History of Ai

DEFINITELY **NOT** AI

DEFINITELY **AI**



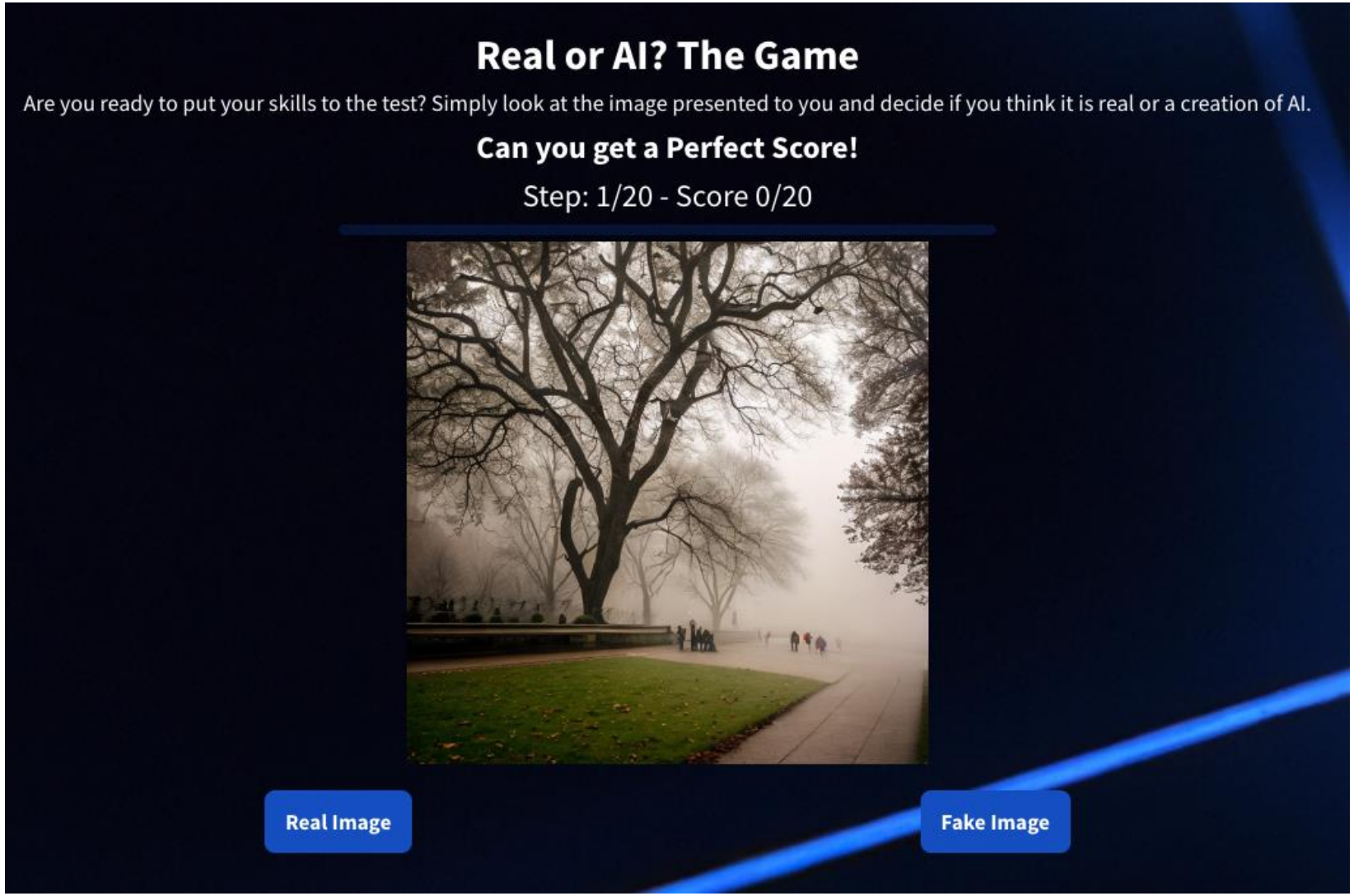
CRAFT
AI LITERACY RESOURCES

Stanford | GRADUATE SCHOOL OF EDUCATION

https://craft.stanford.edu/resource/ai-or-not-ai/



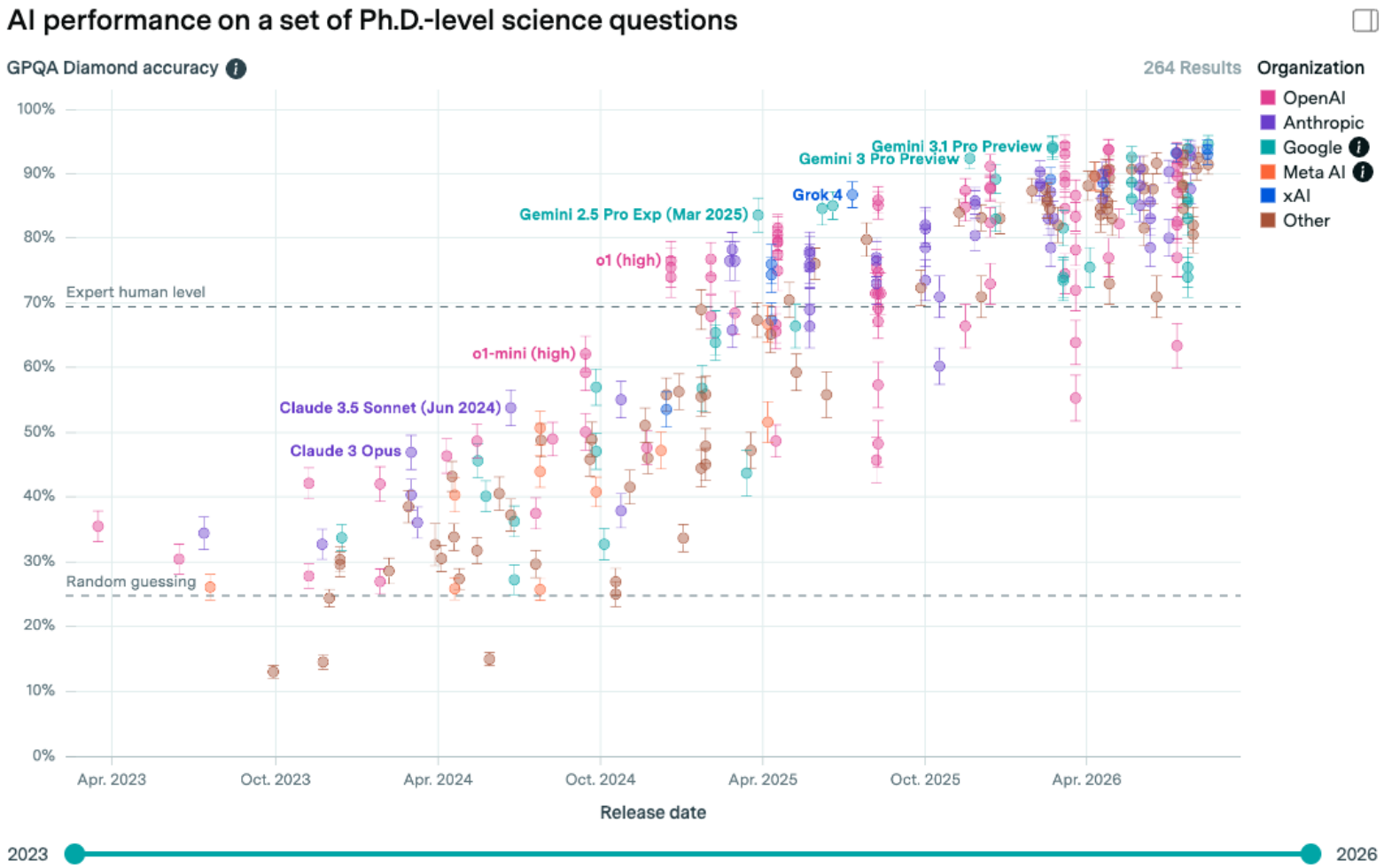
The Canadian House Hippo



Play the “Is it real or is it AI?” Game

https://leonfurze.com/deepfake-game/ https://www.realornotquiz.com/

GPQA – “Google Proof” standardized test for PhD level scientists



https://epoch.ai/benchmarks/gpqa-diamond?view=graph&tab=release-date

Performance of a Large Language Model on the Reasoning tasks of a physician

Editor's summary

Computational tools for medical decision support have been advancing over time, mainly by serving as resources for limited applications. Machine learning tools for autonomous interpretation of clinical cases have also been gradually improving over time. Brodeur *et al*. pitted a large language model, the OpenAI o1 series, directly against hundreds of physicians at different levels of training and experience on a variety of clinical cases ranging from published patient vignettes to evaluations of brand-new emergency room patients, as well as on clinical tasks including both diagnosis and planning of clinical management (see the Perspective by Hopkins and Cornelisse). Across a variety of scenarios and applications, the large language model outperformed both human physicians and older models, suggesting its potential utility for clinical care. —Yevgeniya Nusinovich

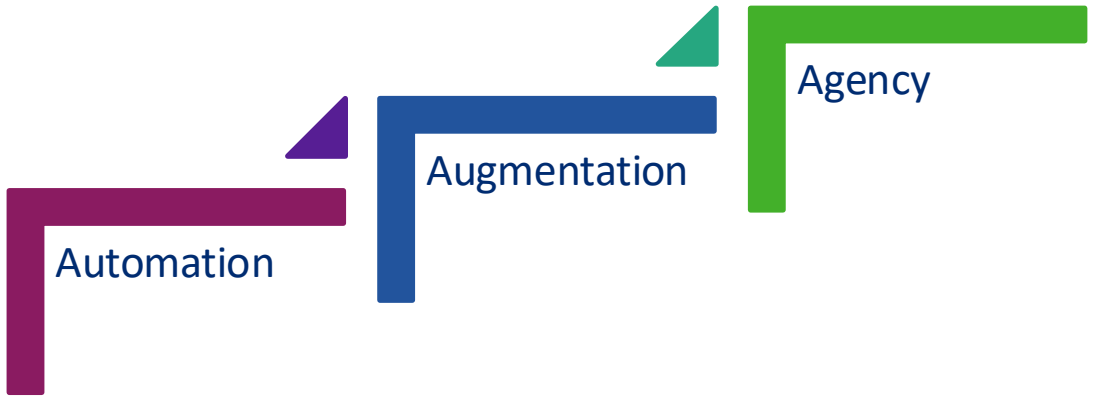
Abstract

More than 65 years ago, complex clinical diagnostic reasoning cases were introduced as the gold standard for the evaluation of expert medical computing systems, a standard that has held ever since. In this study, we report the results of a physician evaluation of a large language model (LLM) on challenging clinical cases across five experiments with a baseline of hundreds of physicians. We then report a real-world study comparing human expert and artificial intelligence (AI) second opinions in randomly selected patients in the emergency room of a major tertiary academic medical center. In all experiments, the LLM outperformed physician baselines and displayed continued improvement from prior generations of AI clinical decision support. Our study suggests that LLMs have eclipsed most benchmarks of clinical reasoning, motivating the urgent need for prospective trials.

https://pubmed.ncbi.nlm.nih.gov/42060751/

Ai Literacy	Ai Fluency
What is Ai? *Understanding the capabilities, limitations, risks, and trustworthiness*	**Engaging with Ai effectively, ethically, safely** *Using Ai to accomplish goals*
• Connection to known tools from Machine Learning, Data Science, and Statistics • What's different?	• **D**elegation • **D**escription • **D**iscernment • **D**iligence

Where should you use "Ai"?



Ai Capabilities & Limitations

What is Ai and how do I work with it?

1. Literacy vs fluency: when to trust, verify, override Ai
2. **What Ai is good and bad at doing**
3. What is *learning*?
4. (Brief) History of Ai

What Ai is surprisingly good at

Capability	Example
Pattern Recognition	Detecting tumors, fraud, anomalies
Language Generation	Summaries, drafting, translation
Information Synthesis	Combining many sources into a coherent overview
Prediction & Classification	Risk scores, recommendations, categorization

Ai is good at finding patterns; humans are still better at deciding what those patterns ***mean***

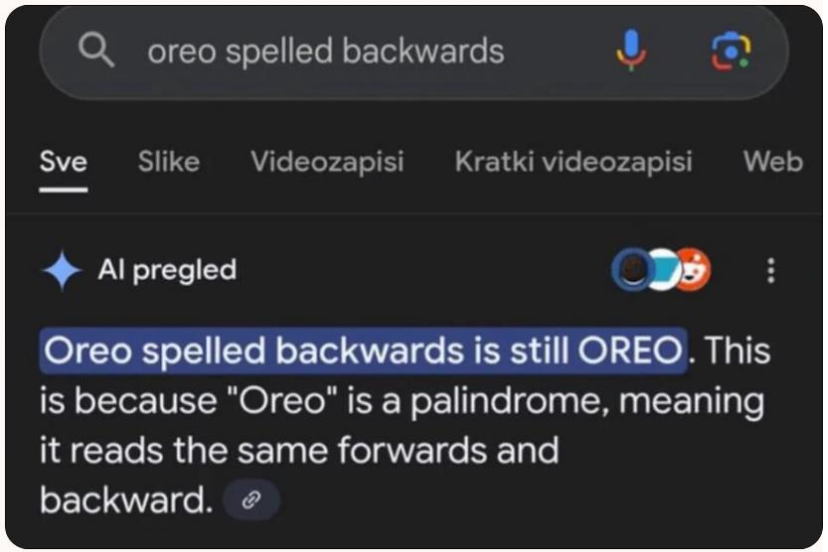
What Ai is surprisingly bad at

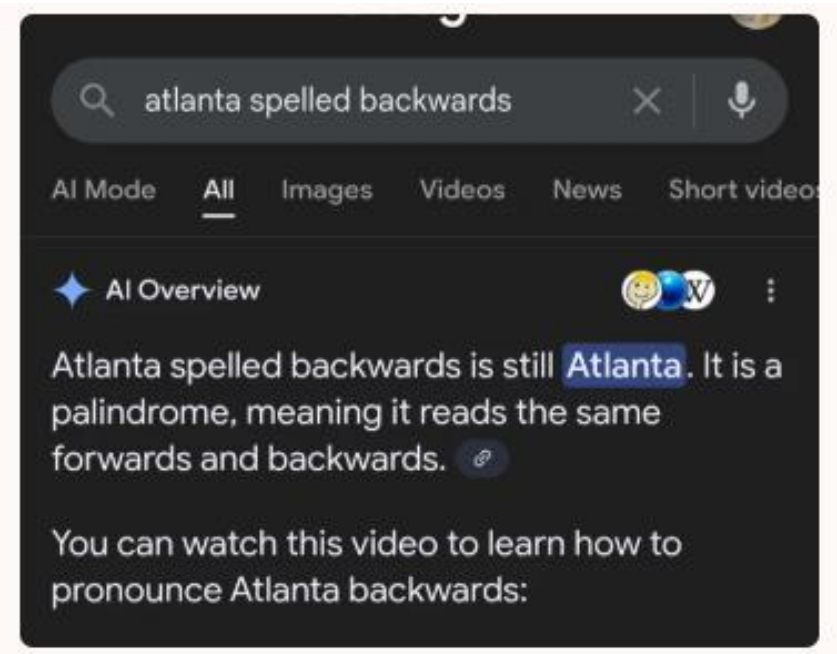
Hallucination
- Confident but false info
- Fluent ≠ factual
- Examples: next slide

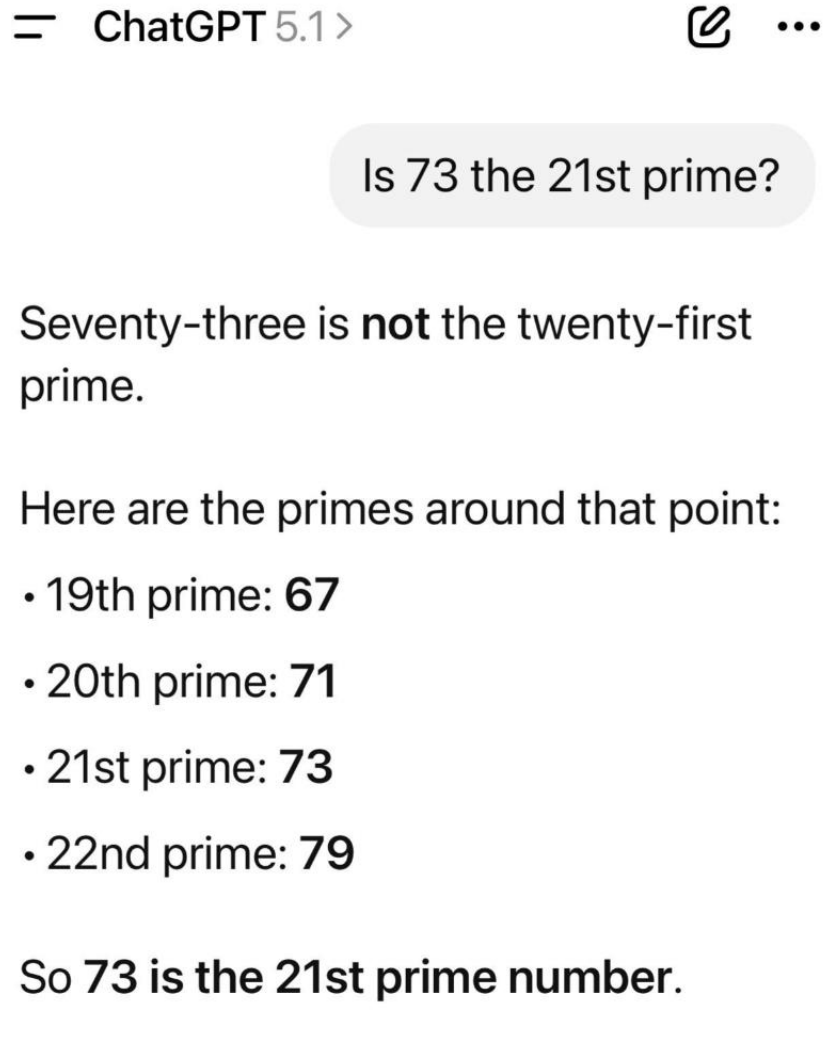
Distribution Shift/Reality changes
- Ai assumes that tomorrow looks like yesterday (or training data)
  - Covid disrupted healthcare models
  - New scientific discoveries
  - Different populations

Brittleness
- ‘intelligence’ is often narrower than it appears
- Small changes can cause BIG failures
- Reworded prompt →new context

Optimization Misalignment
- Ai does what you measure, not what you intend
  - Engagement algorithms
  - Reward hacking







https://www.reddit.com/r/ChatGPT/comments/1pbv54l/73_is_not_the_21st_prime/#lightbox

https://www.reddit.com/r/mildlyinfuriating/comments/1mnakmj/really/

Using Ai (Fluency) versus ***understanding Ai*** (Literacy)

I Can Use AI	I Understand AI
Writes emails	Knows when email content ***should*** be verified
Summarizes papers	Understands summaries ***may*** omit critical details
Generates code	Recognizes code ***may*** contain subtle errors
Answers questions	Knows answers ***may*** be hallucinated

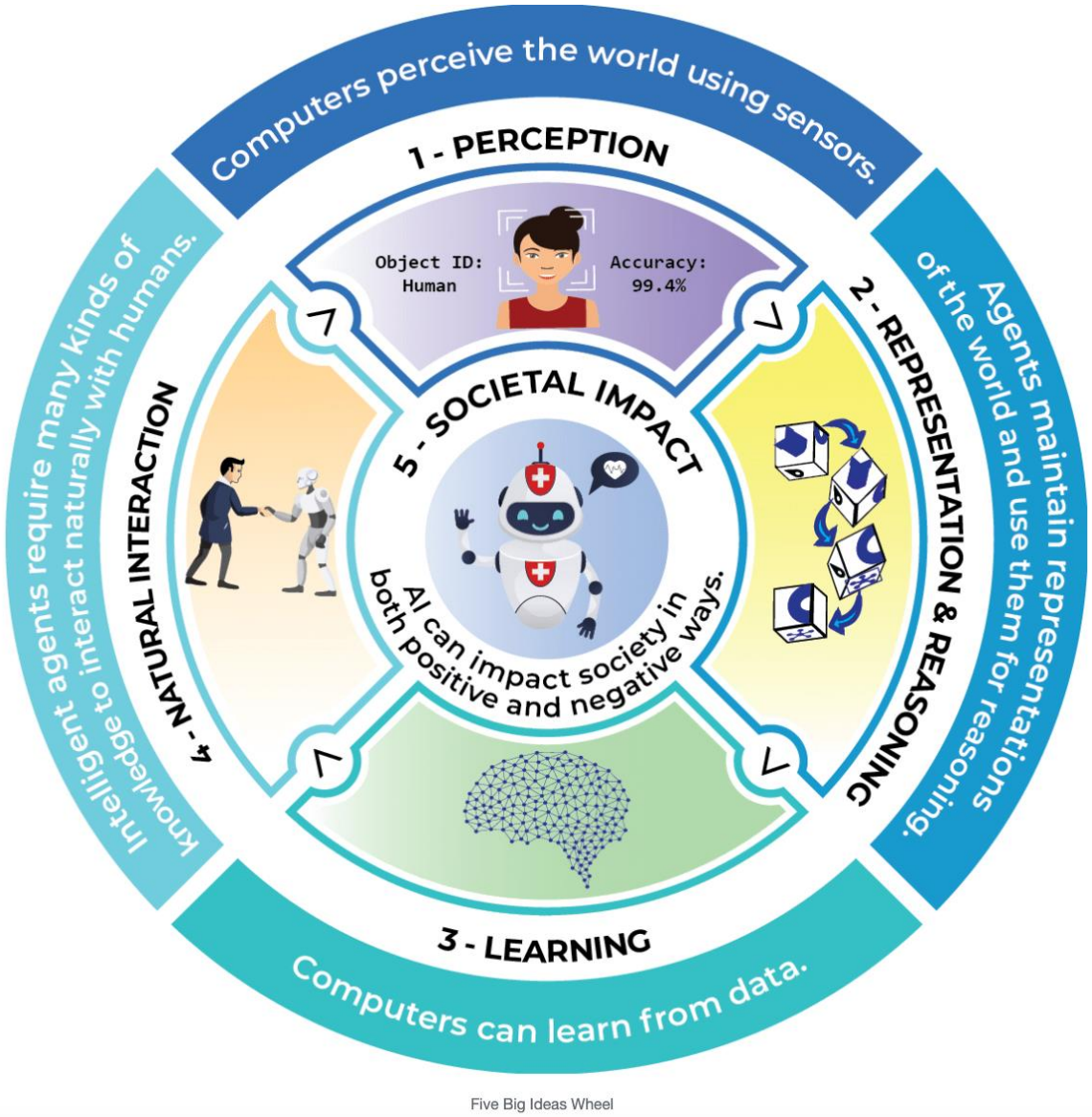
Ai literacy:

- Understanding the capabilities, limitations, risks, and trustworthiness
- **knowing when to trust, verify, reject, or override an Ai system**

Ai Capabilities & Limitations

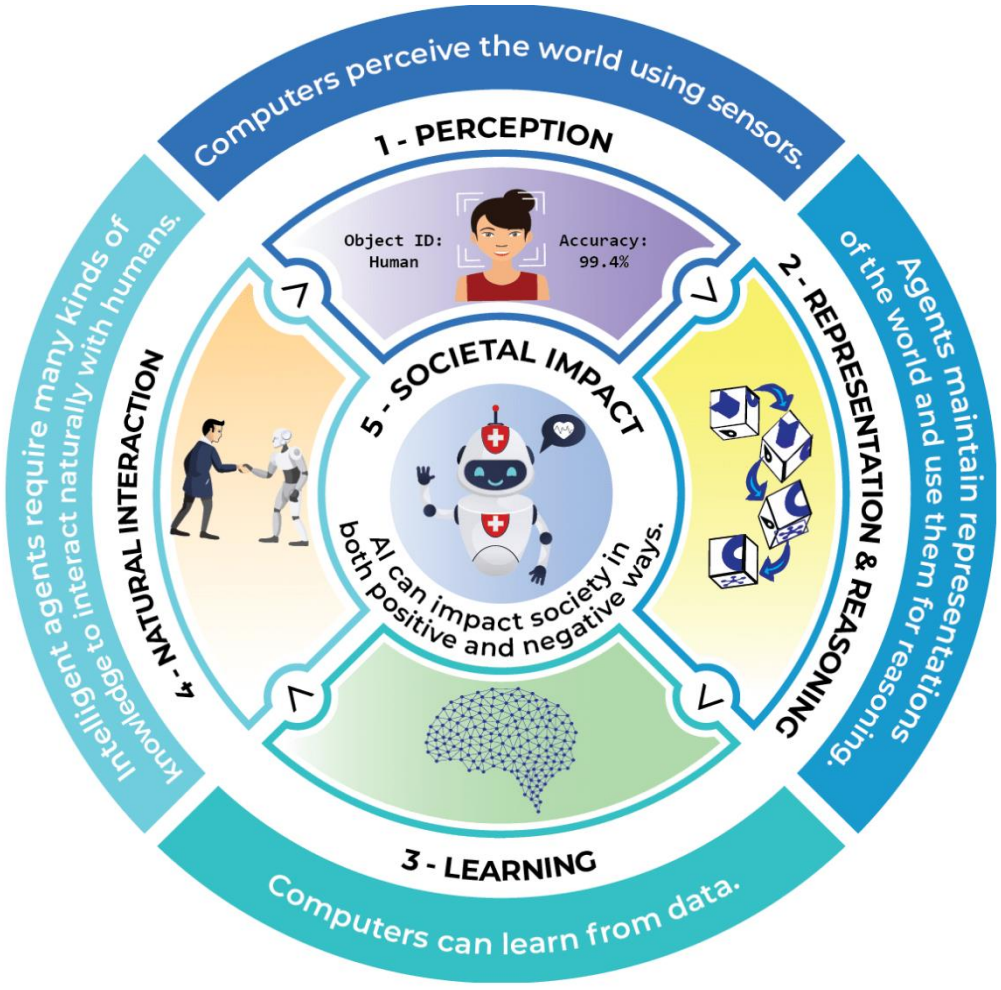
What is Ai and how do I work with it?

1. Literacy vs fluency: when to trust, verify, override Ai
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3. **What is *learning*?**
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Five Big Ideas Wheel

https://ai4k12.org/resources/big-ideas-poster/



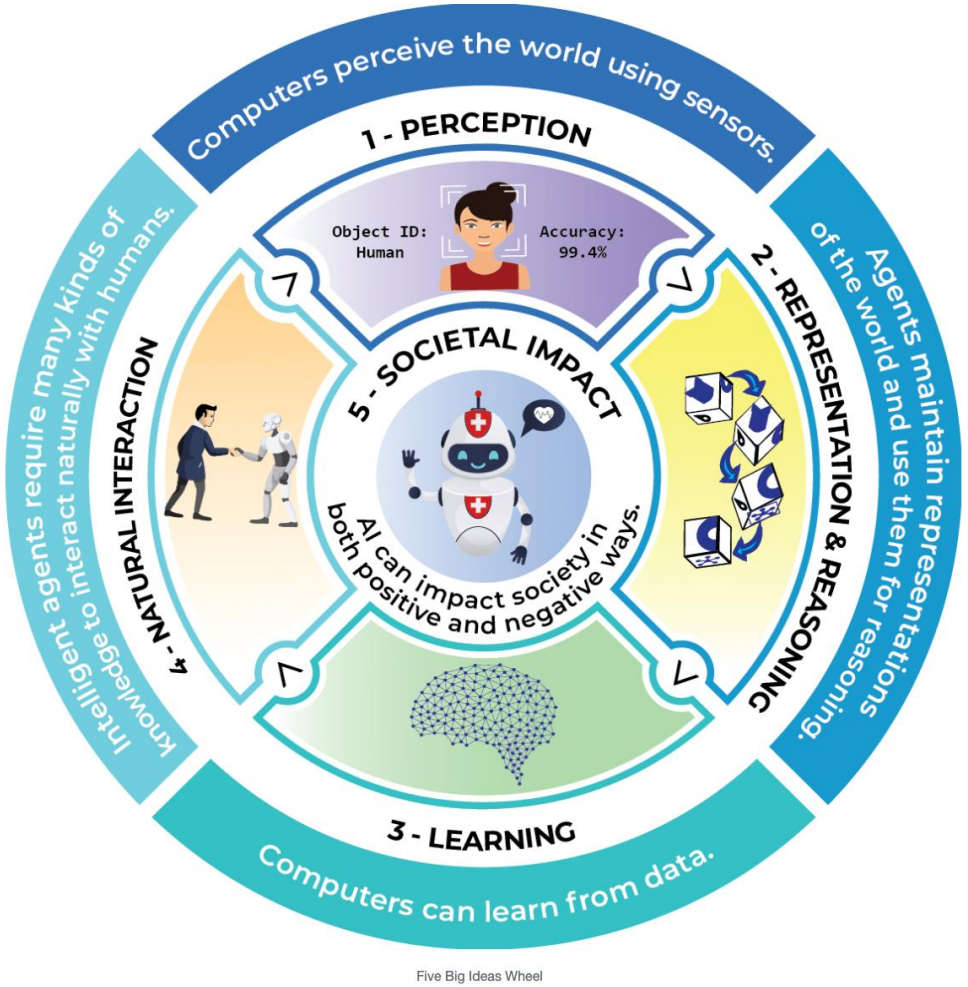
Five Big Ideas Wheel

How do real-world data, like text, images, audio become numerical input?

Ai doesn't *eliminate* the need for thinking clearly,

it amplifies the consequences of thinking *poorly.*

What *is* learning?



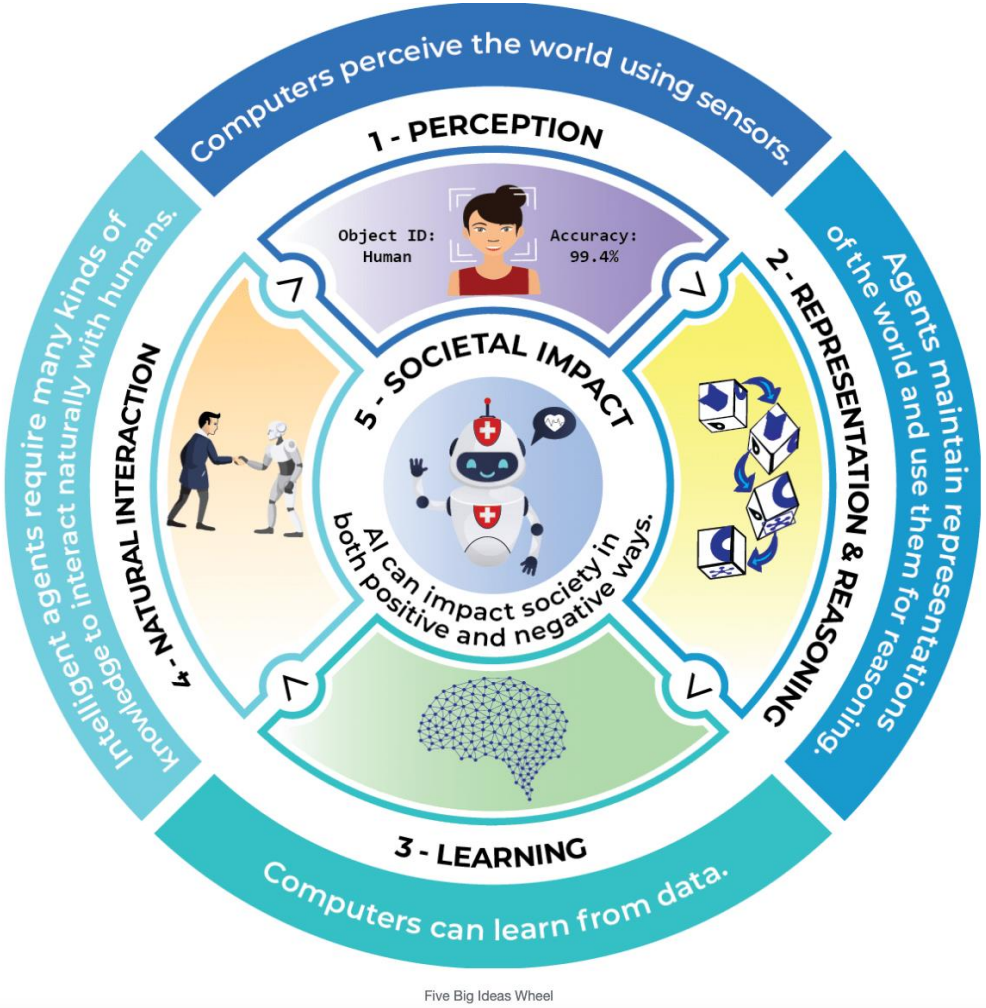
Five Big Ideas Wheel

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AI doesn't *eliminate* the need for thinking clearly, it amplifies the consequences of thinking *poorly*.

What *is* learning?

https://ai4k12.org/resources/big-ideas-poster/



Five Big Ideas Wheel

How do real-world data, like text, images, audio become numerical input?

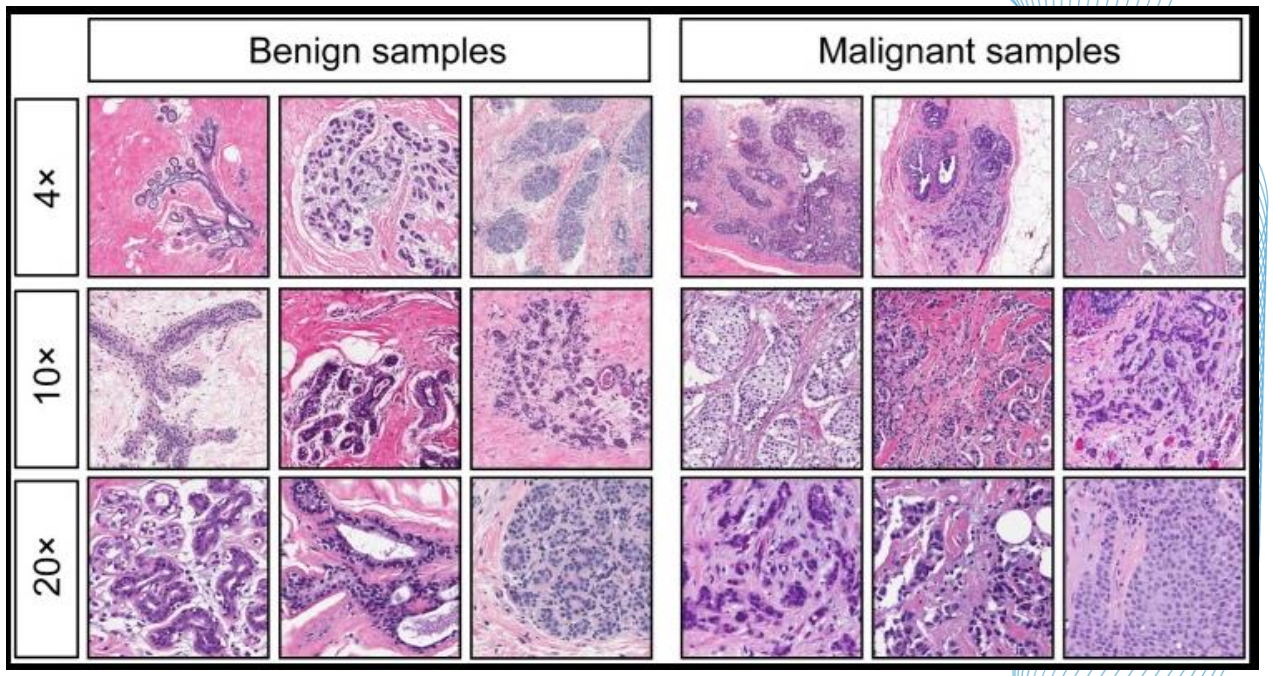
- **Images**→ pixel matrices
- **Text** → tokens → numbers
- **Categories** → one hot encoding

AI doesn't *eliminate* the need for thinking clearly, it amplifies the consequences of thinking *poorly.*

What *is* learning?

What *is* learning?





What *is* learning?



PLOS One

PLoS One. 2015 Nov 18;10(11):e0141357. doi: 10.1371/journal.pone.0141357

Pigeons (*Columba livia*) as Trainable Observers of Pathology and Radiology Breast Cancer Images

Richard M Levenson [1,*], Elizabeth A Krupinski [3], Victor M Navarro [2], Edward A Wasserman [2,*]

Editor: Jonathan A Coles[4]

Author information · Article notes · Copyright and License information

PMCID: PMC4651348 PMID: 26581091

Flock wisdom:

- By day 14, individual birds had 85% accuracy
- A 4 pigeon 'flock' had 99% accuracy → consensus

Home › Learning & Behavior › Article

Taking pigeons to heart: Birds proficiently diagnose human cardiac disease

Published: 21 January 2020

Volume 48, pages 9–21, (2020) Cite this article

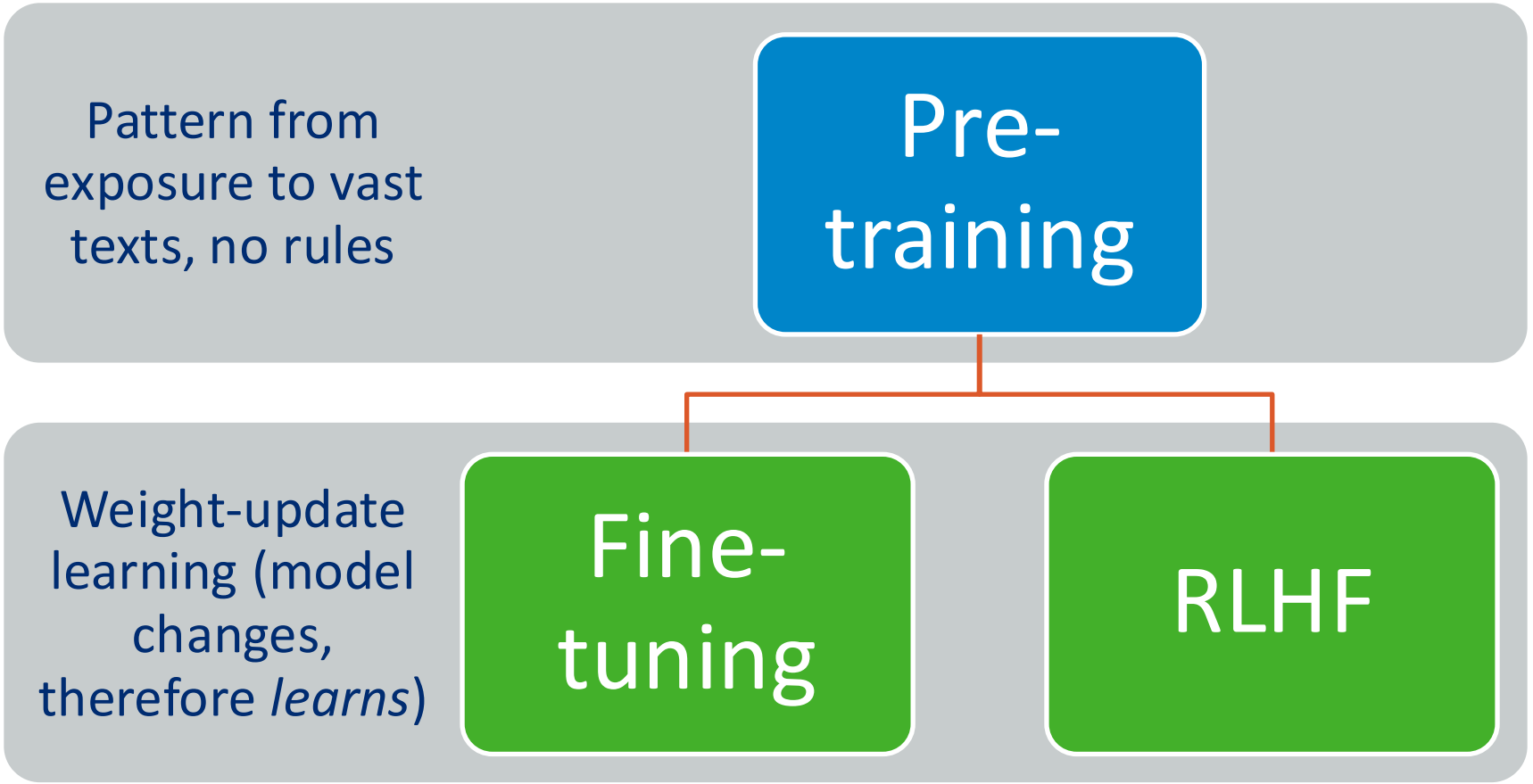
* How many "pigeon units" of training does your physician have?

What *is* learning? The Chinese Room Thought Experiment – John Searle



Conclusion: The person in the room can produce appropriate Chinese responses without understanding a word of Chinese. Therefore, a computer program (which manipulates symbols) is not sufficient for understanding or consciousness.

Types of learning in LLMs



What *is* Learning?

Learning Theory	Human Learning	LLM Equivalent	Where the Analogy Breaks
Behaviorism	Learning through reinforcement, rewards, punishment (**Skinner**)	Reinforcement learning (RLHF/RLAIF) fine-tuning	LLMs receive reinforcement only during training, not continuously while chatting (unless specifically designed to learn online)
Cognitivism	Internal mental representations and information processing **(Piaget)**	Transformer attention, embeddings, internal representations	LLMs lack persistent beliefs, goals, or executive control
Constructivism	Learners actively build knowledge through experience **(Piaget)** and collaboration with more knowledgeable others **(Vygotsky)**	Pretraining builds statistical world models from data	LLMs do not independently experiment or intentionally revise concepts
Social Learning Theory	Learn by observing others (**Bandura**)	Pretraining on human-written text	LLMs observe language but not actual social interactions or intentions
Situated Learning	Knowledge depends on context and environment	In-context learning resembles adapting to local context	LLMs are not situated in the physical world
Experiential Learning	Learning through action and reflection	Limited; tool use and agent systems move closer	LLMs don't learn from experience after deployment (except Agentic systems)
Connectionism	Intelligence emerges from networks of simple units **(Siemens & Downes)**	Probably the closest match	Biological neurons differ enormously from artificial neurons

Learning is relatively permanent change in knowledge or behavior resulting from experience - Wikipedia

Human Learning vs. LLM Learning

HOW HUMANS LEARN	KEY DIMENSION	HOW LLMs LEARN
Continuous & Lifelong Learn from birth through life via ongoing experiences.	When Learning Happens	**Mostly During Training** Learn from large datasets during training. Do not learn (or learn very little) during normal use.
Embodied & Situated Learn through the body and interaction with the physical and social world.	Connection to the World	**Disembodied & Decontextualized** Learn from text data about the world, not direct experience.
Social & Collaborative Learn from others through language, teaching, imitation, and collaboration.	Source of Knowledge	**Observational from Text** Learn from human-written text (books, web, conversations), not live interaction.
Curiosity & Motivation Driven by intrinsic curiosity, emotions, needs, and goals.	Why We Learn	**No Intrinsic Motivation** No goals or curiosity; generates responses based on patterns and instructions.
Learn by Doing & Reflecting Experience → Reflect → Conceptualize → Experiment → Repeat (continual cycle)	How Learning Happens	**No Ongoing Experience Loop** Predicts based on patterns. Does not act in the world or learn from outcomes (unless in agent systems).
Adaptive Memory Memories change and strengthen over time; shaped by emotion and relevance.	Memory	**Fixed (Mostly)** Knowledge is encoded in model weights. Context window is temporary unless explicit memory is added.
Goals & Intentions Learn to achieve goals, solve problems, connect, survive, and grow.	Goals	**Follows Given Objectives** Optimizes next-token prediction and follows user or training objectives.

SIMILARITIES

- Both build internal representations of patterns.
- Both improve by seeing many examples.
- Both use prediction and update based on error.

KEY DIFFERENCES

- Humans are embodied, emotionally driven, and socially interactive.
- Humans learn continuously from experience.
- LLMs learn mostly during training from text data.
- LLMs lack true curiosity, intentions, and consciousness.

Bottom Line Humans learn through a lifetime of embodied, social, goal-directed experiences.
LLMs learn by compressing patterns from vast amounts of text during training and generate responses based on those patterns.

If an LLM can solve a problem without “understanding’ what does understanding actually mean?

What is *thinking?*

cogito, ergo sum.

What is *thinking?*

***Dubito*, ergo cogito, ergo sum.**

Ai Capabilities & Limitations

What is Ai and how do I work with it?

1. Literacy vs fluency: when to trust, verify, override Ai
2. What Ai is good and bad at doing
3. What is *learning*?
4. **(Brief) History of Ai**

Is Artificial intelligence *new?*

Key moments

1943 **McCullouch and Pitts**

- A Logical Calculus of the Ideas Immanent in Nervous Activity
- laid foundations for “artificial neurons”

1948 **Claude Shannon**

- A Mathematical Theory of Communication

1949 **Hebb**

- “fire together, wire together”

1950 **Turing**

-“Computing Machinery and Intelligence”

1956 **McCarthy**

- Coined the term: Artificial Intelligence
- thinking involves both knowledge and reasoning and can be encoded mathematically

1957 **Rosenblatt**

perceptrons → Neural Networks

1967 **Minsky**

2009 **Fei-Fei Li**

- ImageNet (14 million images labeled)

2012 **AlexNet**

2024 **Nobel Prizes**

- **Physics:** Geoffrey Hinton, John J. Hopfield

1308 Catalan poet and theologian Ramon Llull publishes *Ars generalis ultima* (The Ultimate General Art), further perfecting his method of using paper-based mechanical means to create new knowledge from combinations of concepts.

1666 Mathematician and philosopher Gottfried Leibniz publishes *Dissertatio de arte combinatoria* (On the Combinatorial Art), following Ramon Llull in proposing an alphabet of human thought and arguing that all ideas are nothing but combinations of a relatively small number of simple concepts.

1726 Jonathan Swift publishes *Gulliver's Travels*, which includes a description of the Engine, a machine on the island of Laputa (and a parody of Llull's ideas): "a Project for improving speculative Knowledge by practical and mechanical Operations." By using this "Contrivance," "the most ignorant Person at a reasonable Charge, and with a little bodily Labour, may write Books in Philosophy, Poetry, Politicks, Law, Mathematicks, and Theology, with the least Assistance from Genius or study."

1763 Thomas Bayes develops a framework for reasoning about the probability of events. Bayesian inference will become a leading approach in machine learning.

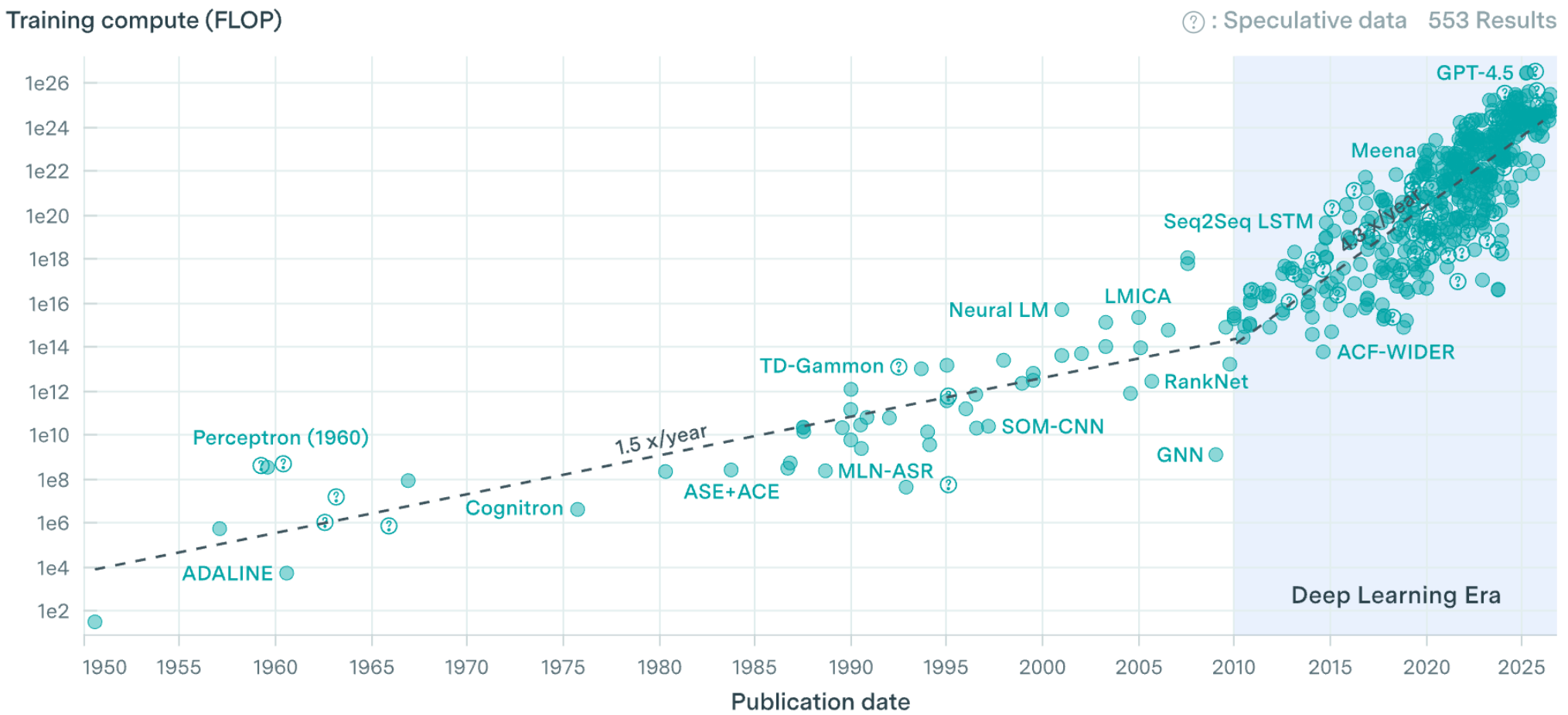
1854 George Boole argues that logical reasoning could be performed systematically in the same manner as solving a system of equations.

1898 At an electrical exhibition in the recently completed Madison Square Garden, Nikola Tesla makes a demonstration of the world's first radio-controlled vessel. The boat was equipped with, as Tesla described, "a borrowed mind."

1914 The Spanish engineer Leonardo Torres y Quevedo demonstrates the first chess-playing machine, capable of king and rook against king endgames without any human intervention.

https://www.tableau.com/data-insights/ai/history

Notable AI models



EPOCH AI | CC-BY

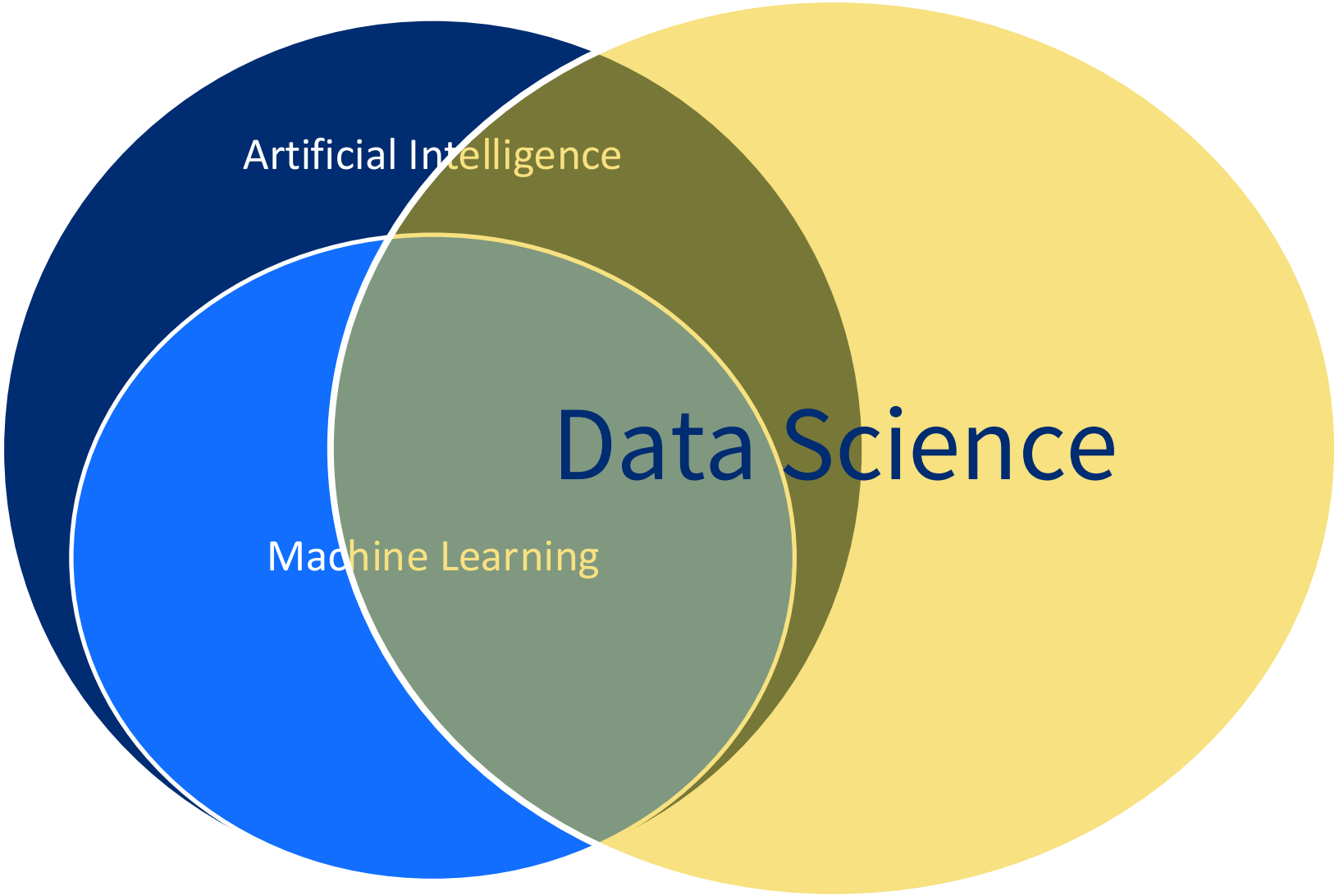
epoch.ai

https://epoch.ai/data/notable-ai-models?view=table#explore-the-data

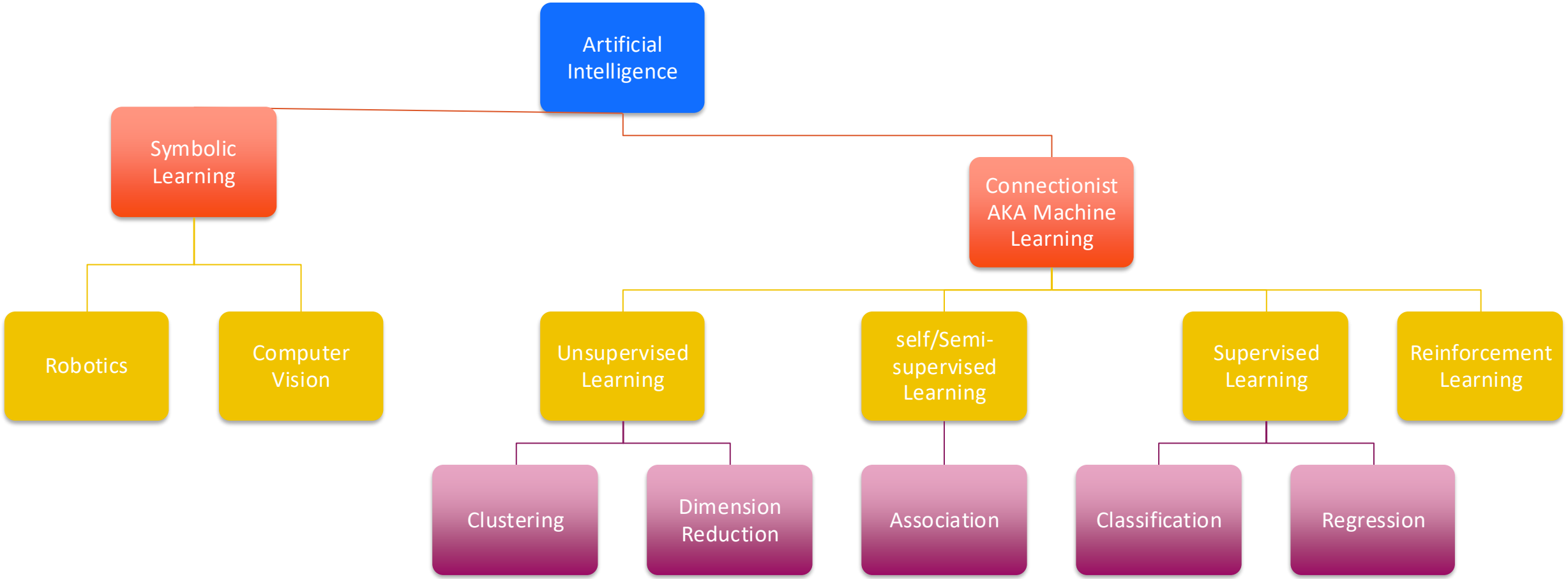
15-minute Break

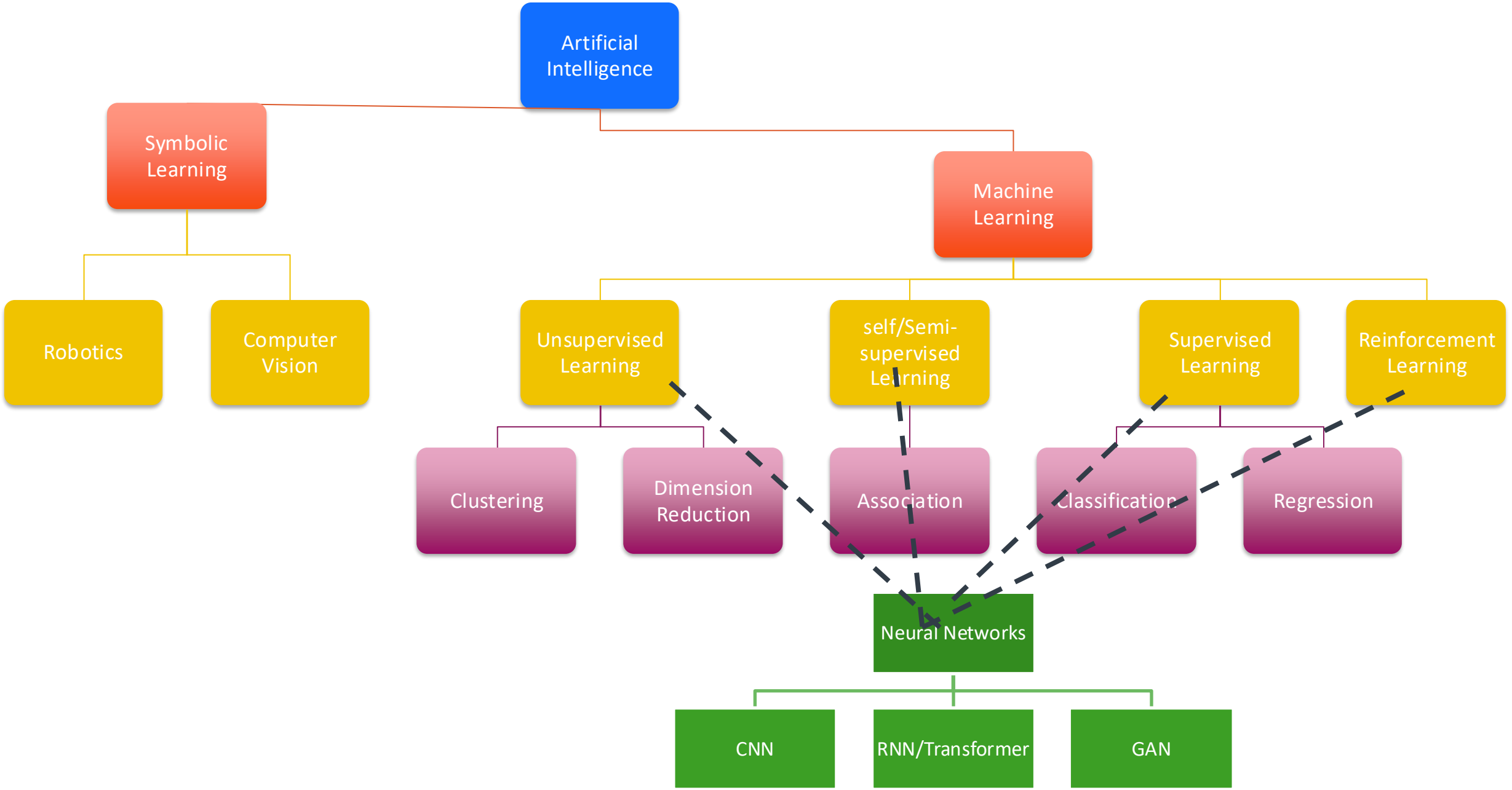
How are Ai, Machine Learning, and Data Science Related?

- Definitions
- Breakdown of Ai into ML categories



Data Science = Statistics + Linear Algebra + Computer Science + Domain Data


Artificial Intelligence
Symbolic Learning
Connectionist AKA Machine Learning
Robotics
Computer Vision
Unsupervised Learning
self/Semi-supervised Learning
Supervised Learning
Reinforcement Learning
Clustering
Dimension Reduction
Association
Classification
Regression


Artificial Intelligence
Symbolic Learning
Machine Learning
Robotics
Computer Vision
Unsupervised Learning
self/Semi-supervised Learning
Supervised Learning
Reinforcement Learning
Clustering
Dimension Reduction
Association
Classification
Regression
Neural Networks
CNN
RNN/Transformer
GAN

- Not necessarily **predictive**
- Group and interpret data based on **input**

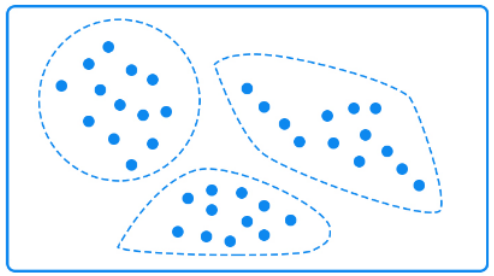
Develop **predictive model** based on **input** and **output**

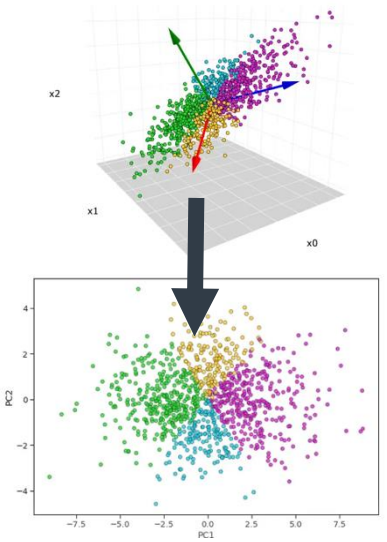
Data is labelled



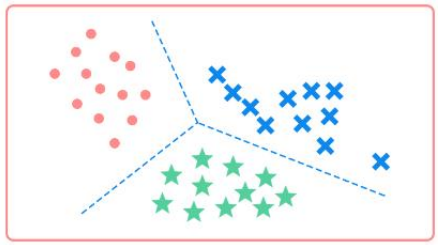


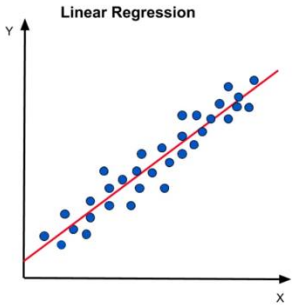


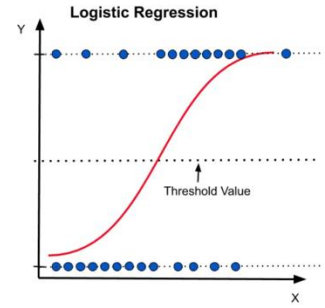


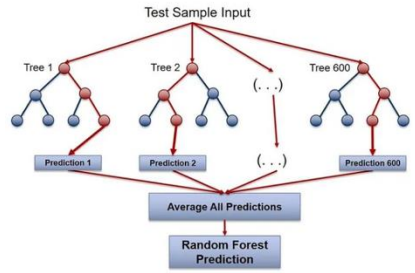










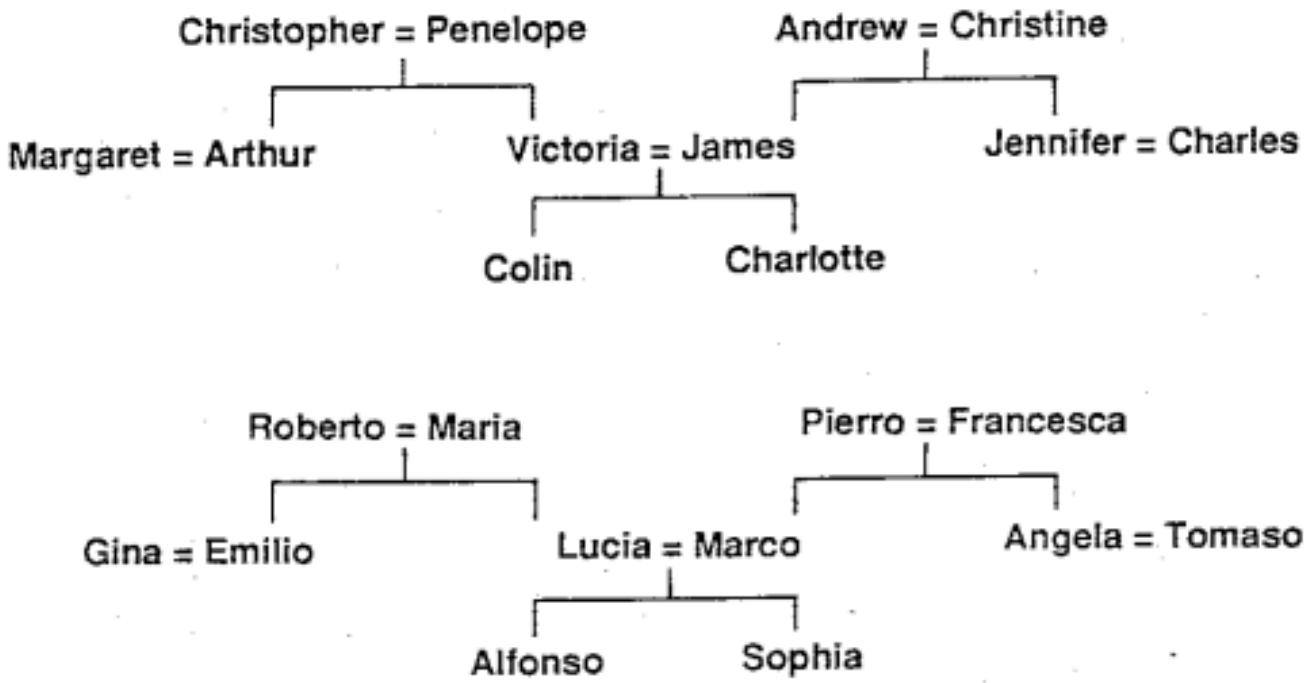


The Features

How do we translate the world into computationally tractable features?

Geoffrey Hinton, 1986 : Learning Distributed Representations of Concepts

Activity: We want to teach a computer about these people. What information would you want the computer to know about each person to predict their family relationships? Find **5** features.



https://www.cs.toronto.edu/~hinton/absps/families.pdf

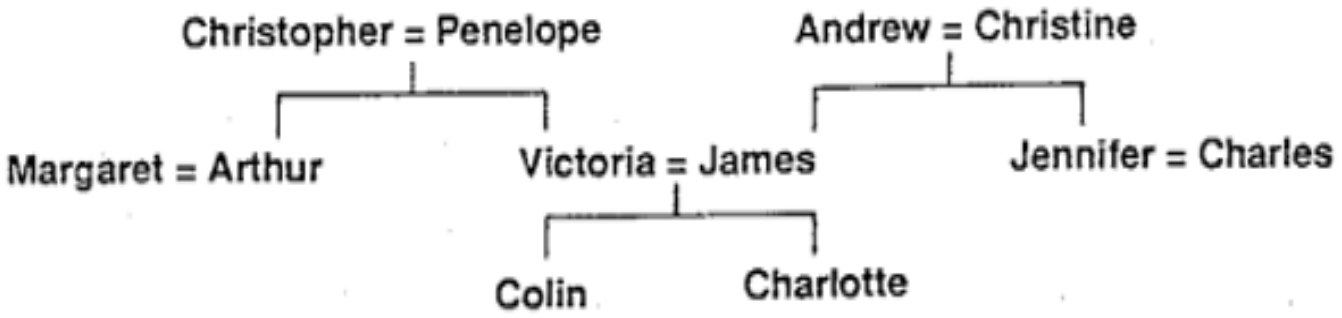
Geoffrey Hinton, 1986 : Learning Distributed Representations of Concepts

Activity: We want to teach a computer about these people. What information would you want the computer to know about each person to predict their family relationships? Find **5** features.

If **Victoria** and **Jennifer** have the same generation and are both female, why aren't they interchangeable?

Person	Generation	Female	Parent	Has Sibling
Victoria	2	1	1	1
James	2	0	1	1
Colin	3	0	0	1
Jennifer	2	1	0	1

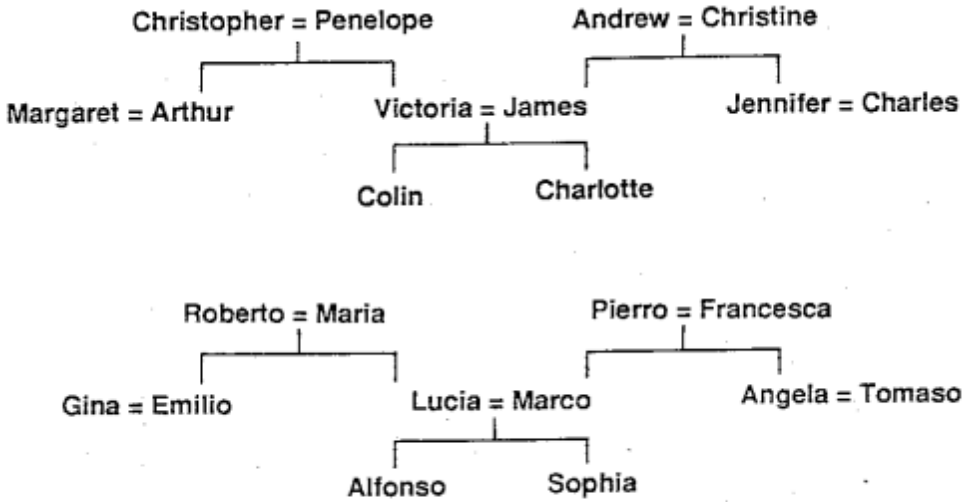
These features are insufficient;
we need to know ***how the people are connected***



https://www.cs.toronto.edu/~hinton/absps/families.pdf

Geoffrey Hinton, 1986 : Learning Distributed Representations of Concepts

Activity: The two families have identical structures. Find the Italian equivalent to each English Person. That is: What do Christopher and Roberto have in common? What do Colin and Alfonso have in common?



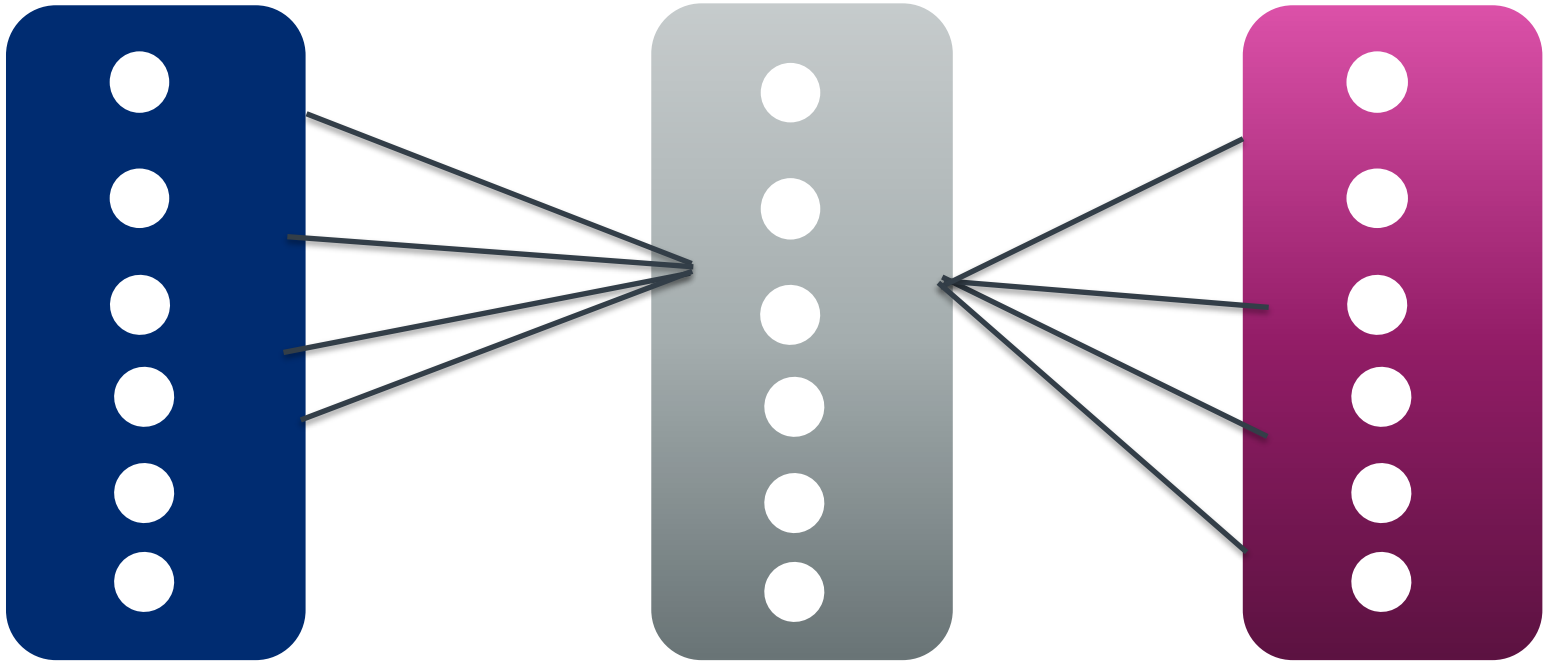
Christopher	→	
Victoria	→	
Jennifer	→	
Colin	→	
Charlotte	→	

Questions:

1. Suppose the network has learned the relationship between Colin and Jennifer. What should it predict about Alfonso and Angela?
2. How could the computer discover that Colin and Alfonso are similar when we never explicitly told it that they were equivalent?

https://www.cs.toronto.edu/~hinton/absps/families.pdf

Geoffrey Hinton, 1986 : Learning Distributed Representations of Concepts



Inputs
Person + relation

Hidden Layer
Learned Features

Prediction
Related Person

https://www.cs.toronto.edu/~hinton/absps/families.pdf

Geoffrey Hinton, 1986 : Learning Distributed Representations of Concepts

Activity: Contrast what the network did with what you did

Participants Did	Hinton Did
Selected features consciously	Learned internal representations from examples
Assigned interpretable labels	Used patterns of numerical activations
Applied kinship rules	Adjusted connection weights to improve predictions
Identified corresponding people	Developed similar representations for structurally equivalent people

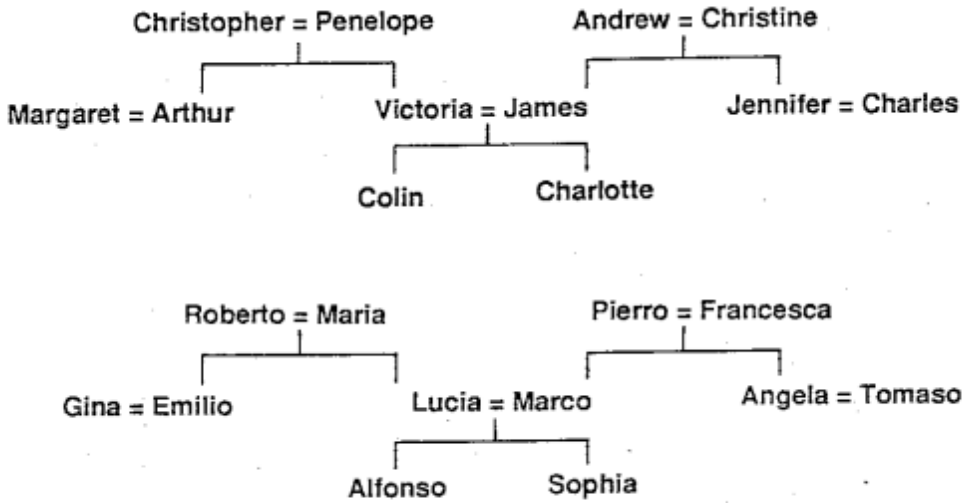


Figure 2: Two isomorphic family trees. The symbol "=" means "married to".

https://www.cs.toronto.edu/~hinton/absps/families.pdf

Geoffrey Hinton, 1986

LEARNING DISTRIBUTED REPRESENTATIONS OF CONCEPTS

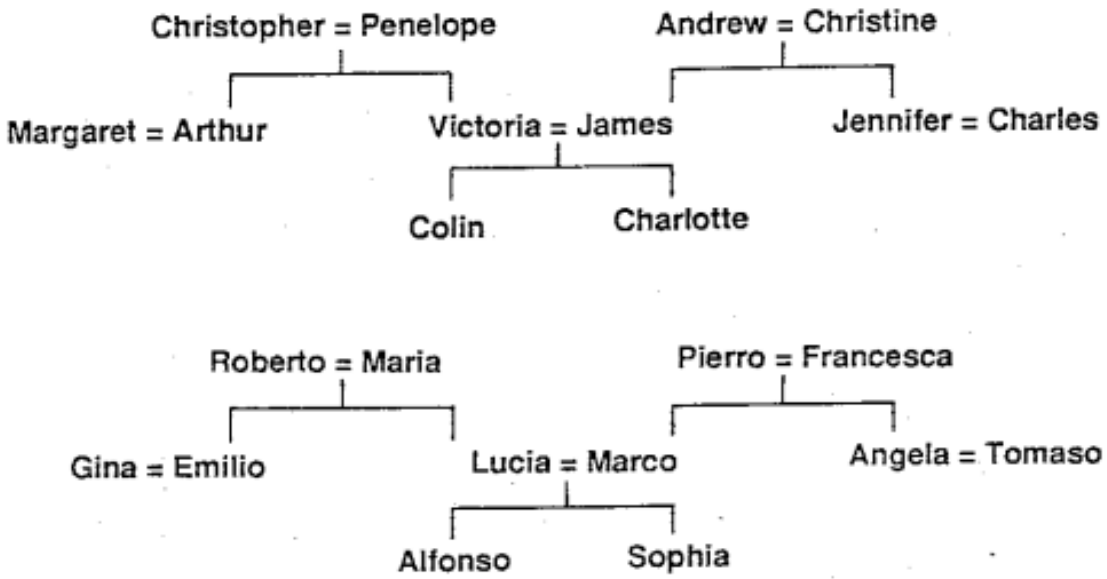
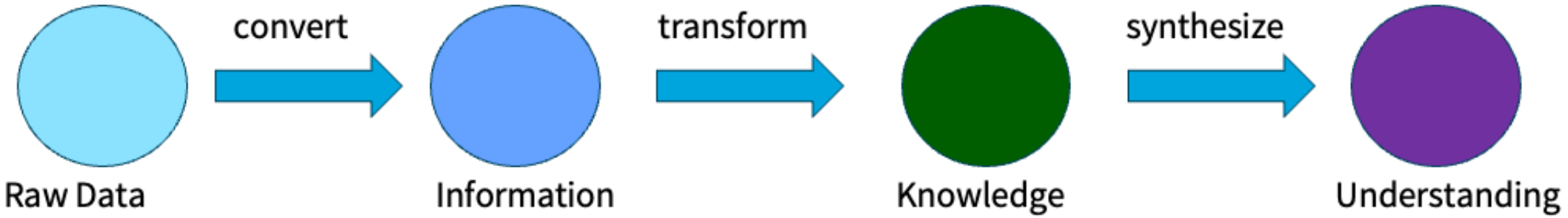


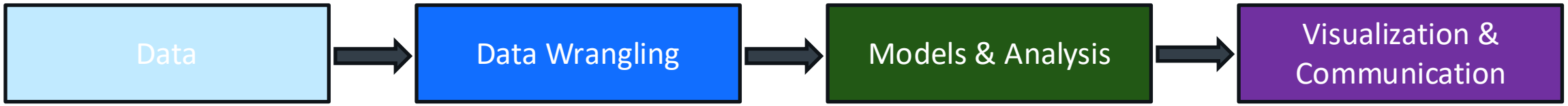
Figure 2: Two isomorphic family trees. The symbol "=" means "married to".

- How to learn a set of semantic features for each word
- **Relational knowledge** resides in the way that the features **interact**, not in static propositions
- Make a set of triples using 12 relationships
  - Son, daughter, nephew, niece
  - Father, mother, uncle, aunt
  - Brother, sister, husband, wife
  - (Colin has-father James)
  - (Colin has-mother Victoria)
  - (James has-wife Victoria)
  - (Charlotte has-brother Colin)
  - (Victoria has-brother Arthur) etc

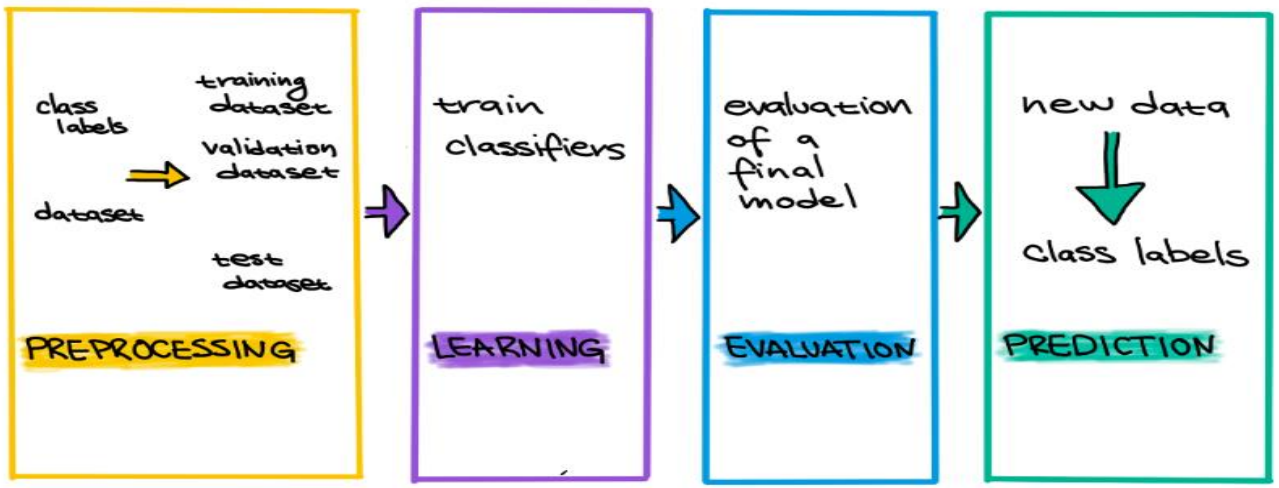
These are ISOMORPHIC trees. *Has the algorithm learned kinship relationships?*

https://www.cs.toronto.edu/~hinton/absps/families.pdf





Machine learning workflow



Raw Data Becomes Numbers:

- All inputs – pixels, words, sounds – must be converted into numbers so that models can process them

Images

Pixels→ RGB values → Matrices

Categories

Labels→ One hot Encoding

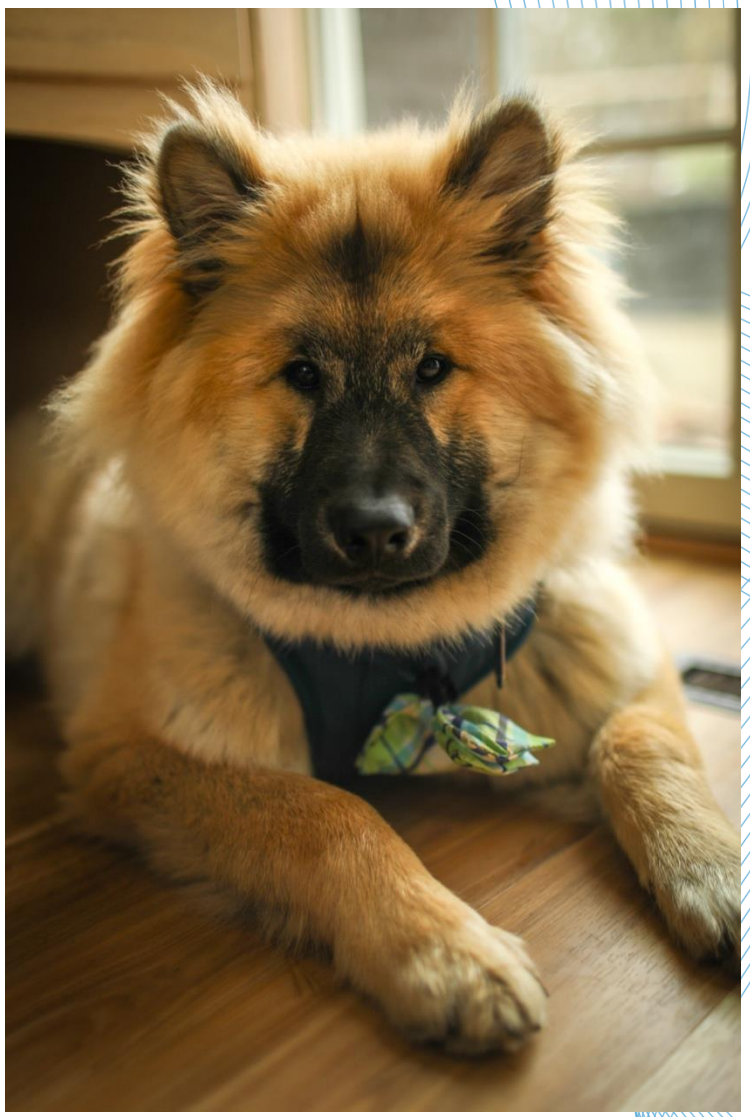
Text

Words→ Tokens → Embeddings

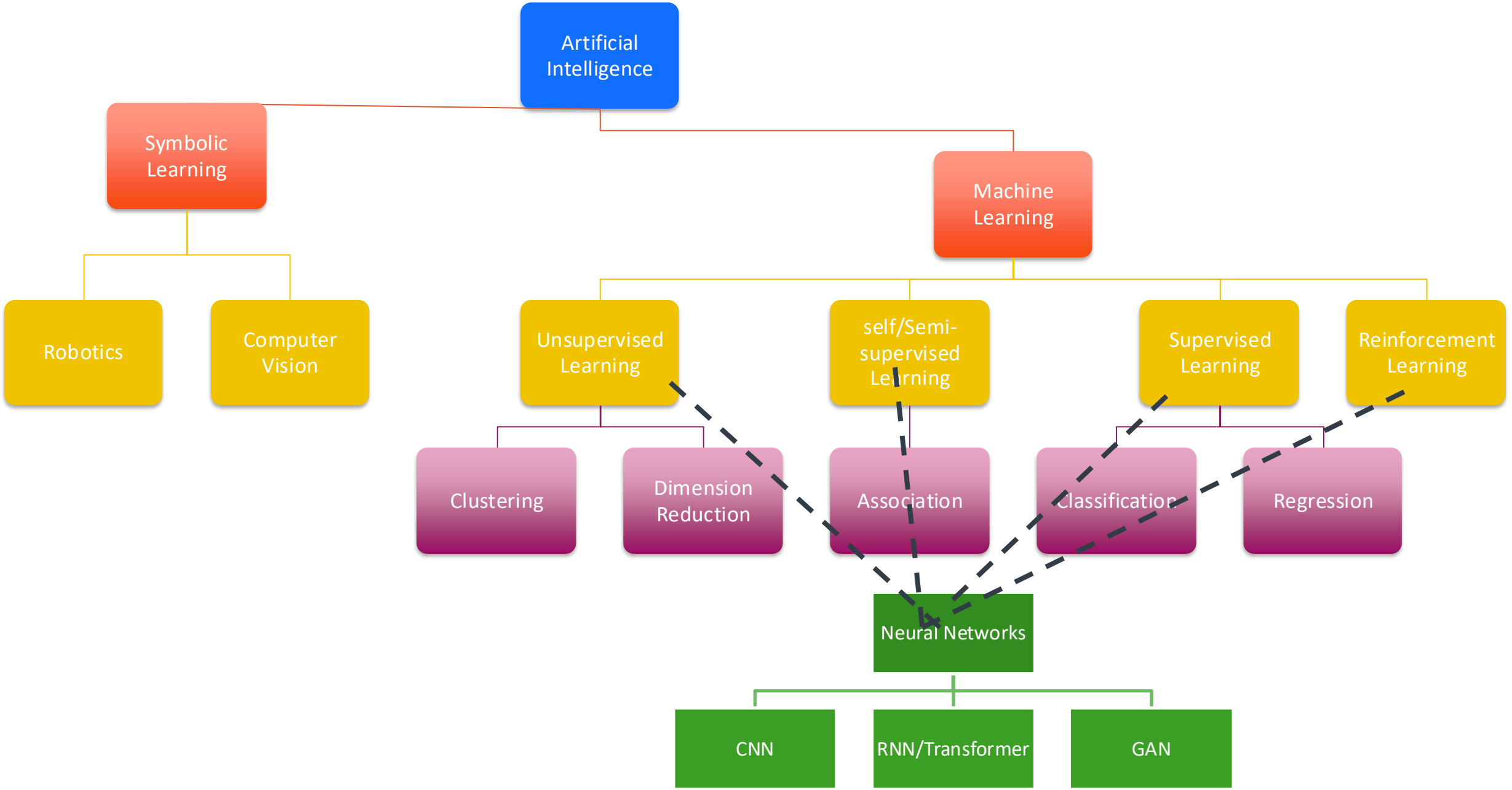
Audio

Waveforms→ Frequencies → Vectors

Huxley is an 18-month-old Eurasier (male, not neutered) who came in on 110/03/2025 for a sick visit because his owner said he hasn't been eating normally and vomited twice. The tech wrote his weight as **40 pounds**, though later in the notes someone jotted "~18 kg?" with a question mark next to it. On exam, Huxely's temperature was 102.1 F, heart rate 96 bpm, and his respiratory rate was listed as "**about 30/min**," although the structured intake sheet earlier said 28/min. His body condition score was 6/9 and hydration was estimated around 5% dehydrated. Pain was mild (1 on the 0–4 scale). CBC showed WBC 14.2 (slightly high), RBC 6.3, hemoglobin 15.1 g/dL, and platelets 268. Chemistry panel had BUN 21, creatinine 1.1, ALT 68 (a bit high), and glucose 87. Abdominal radiographs noted some gastric fluid but no foreign body. The assessment was possible gastritis with mild dehydration, no obstruction suspected. Plan: maropitant at 1 mg/kg once daily for 3 days, 200 mL SQ fluids, bland diet for 2–3 days with small frequent meals and return if vomiting continues for more than 48 hours.



15-minute Break


Artificial Intelligence
Symbolic Learning
Machine Learning
Robotics
Computer Vision
Unsupervised Learning
self/Semi-supervised Learning
Supervised Learning
Reinforcement Learning
Clustering
Dimension Reduction
Association
Classification
Regression
Neural Networks
CNN
RNN/Transformer
GAN

Machine Learning

Unsupervised Learning

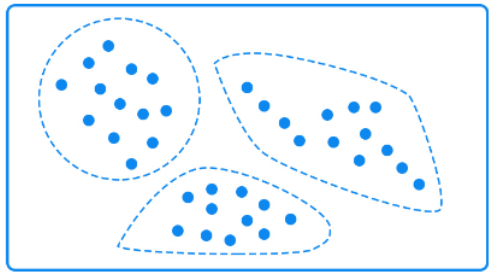
Supervised Learning

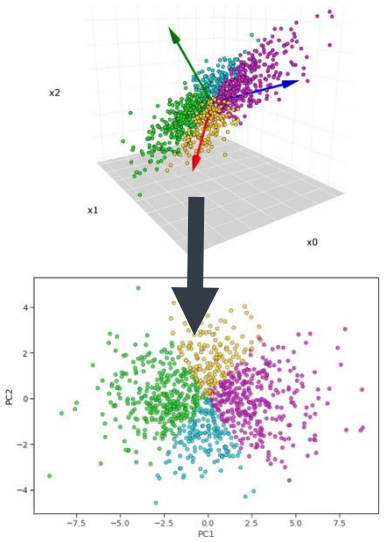
Clustering

Dimension Reduction

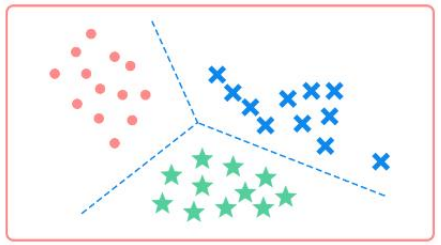
Classification

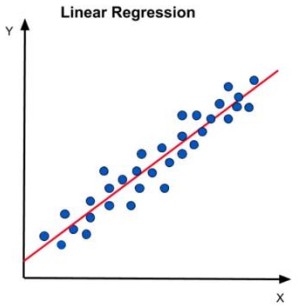
Regression

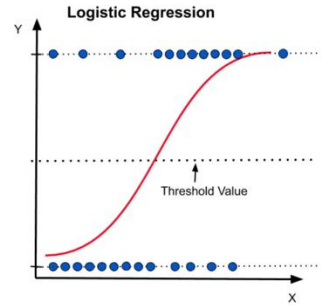


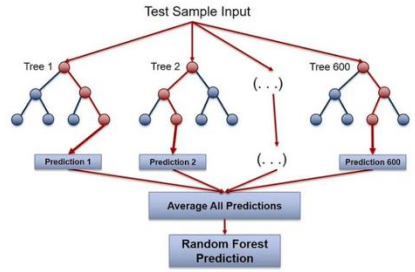










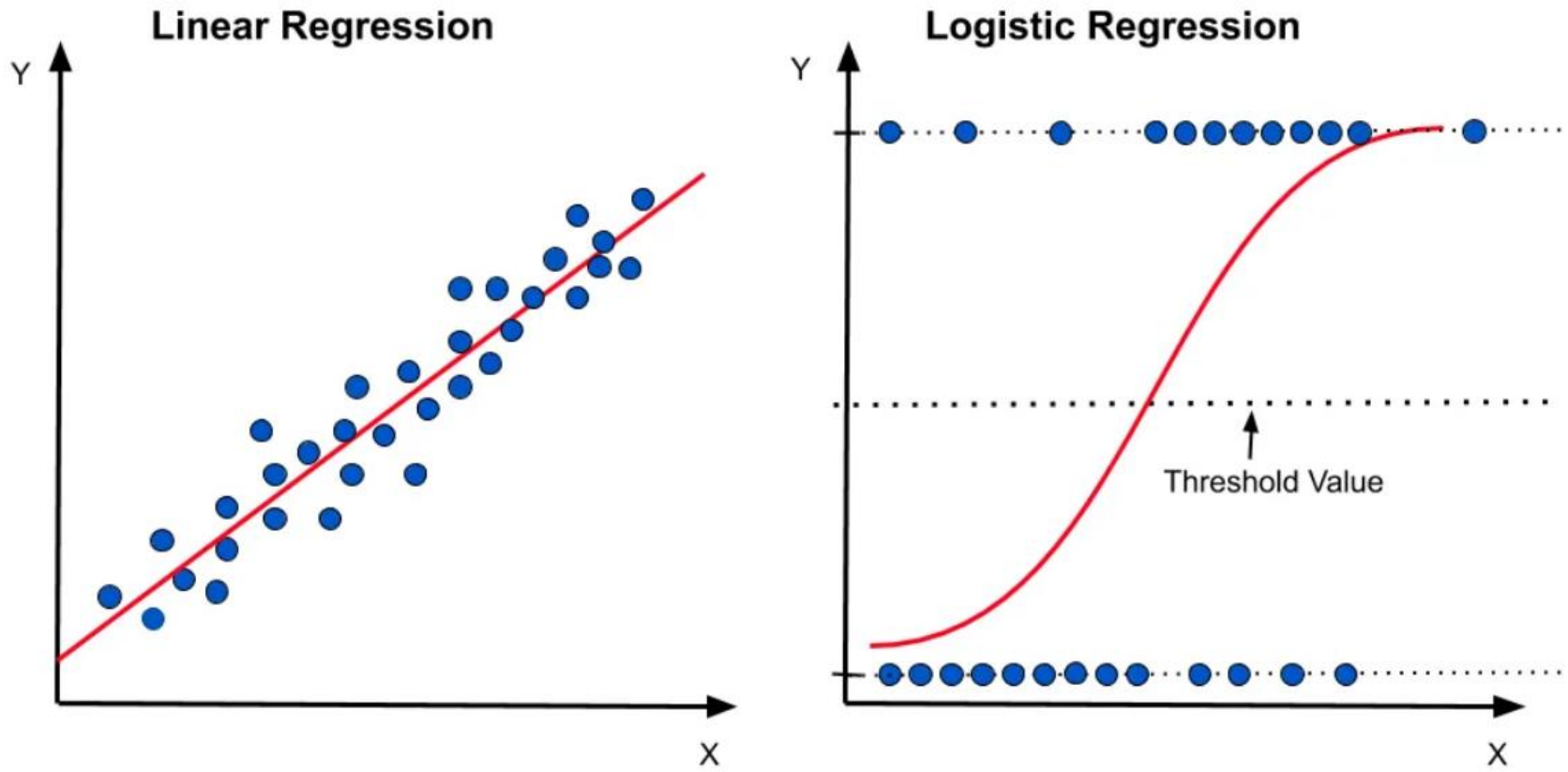


How machines learn: Regression, K-means & Loss Functions

Building intuition for the machinery of everything that we will discuss this week

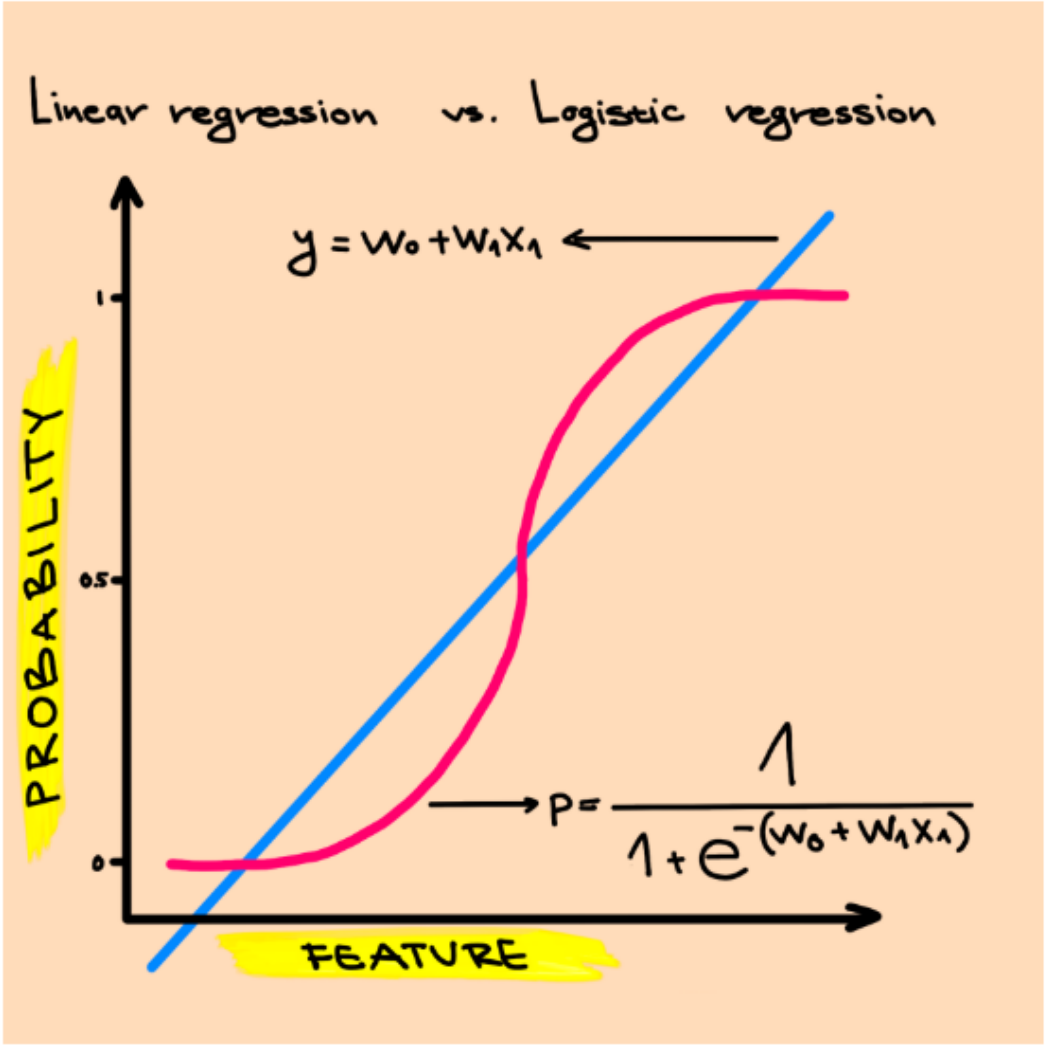
In machine learning, computers learn patterns that help them to make decisions without being explicitly programmed.

- A familiar way to make predictions is with regression:



For decision trees example: http://www.r2d3.us/visual-intro-to-machine-learning-part-1/

Supervised ML: **Logistic & Linear Regression**



https://carpentries-incubator.github.io/ml4bio-workshop/05-logit-ann/index.html

Questions to consider for Logistic Regression:

- **How** do Ai systems use features (input variables) to generate predictions?
- **How** do supervised models make decisions?
- Can we **impact** the decision that it makes?
- Consider **where** logistic regression models are used (healthcare, criminal justice, finance).
  - What are the risks of defining the cut-off value?
  - Of including certain features and not others?

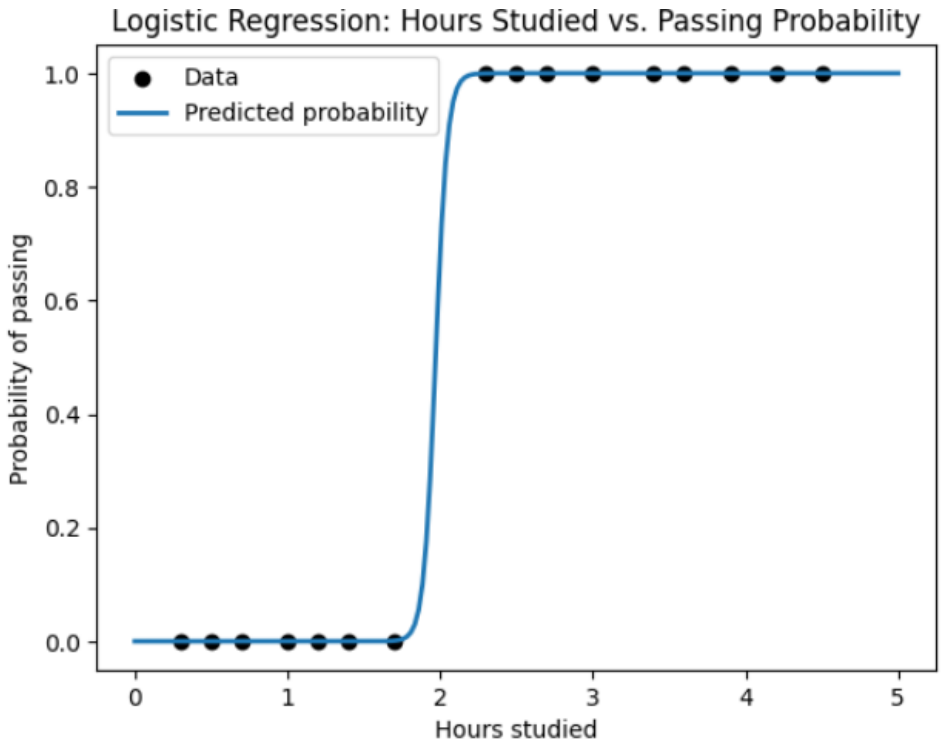
Supervised ML: Logistic Regression

Hours studied	Result
0.3, 0.5, 0.7, 1.0, 1.2, 1.4, 1.7	Fail
2.3, 2.5, 2.7, 3.0, 3.4, 3.6, 3.9, 4.2, 4.5	Pass

Modified example from Wikipedia page on Logistic Regression

Supervised ML: Logistic Regression

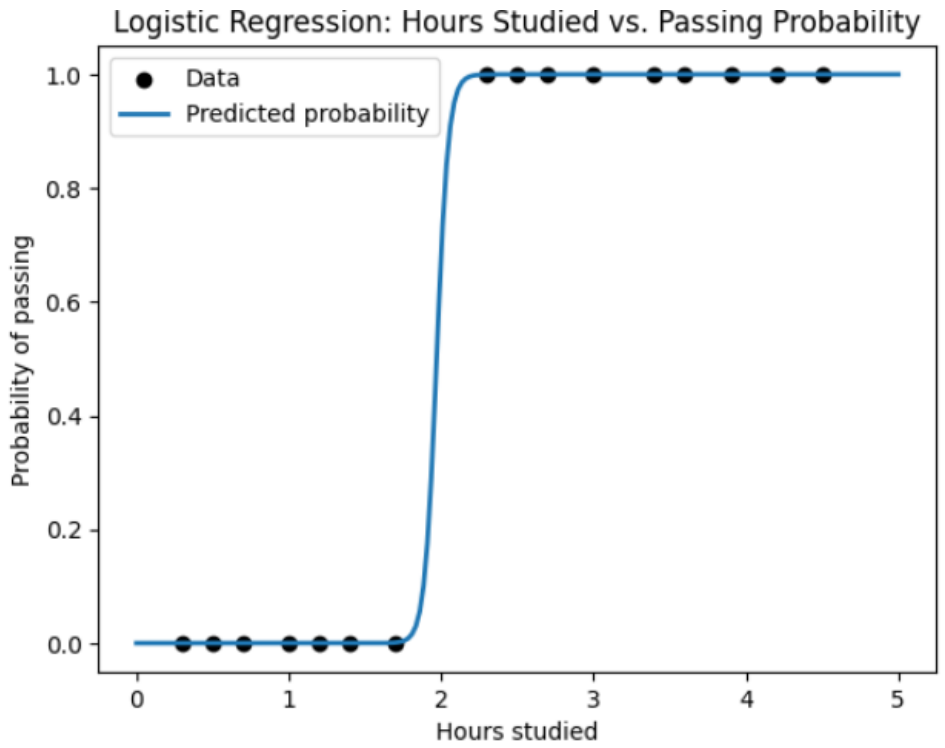
Hours studied	Result
0.3, 0.5, 0.7, 1.0, 1.2, 1.4, 1.7	Fail
2.3, 2.5, 2.7, 3.0, 3.4, 3.6, 3.9, 4.2, 4.5	Pass

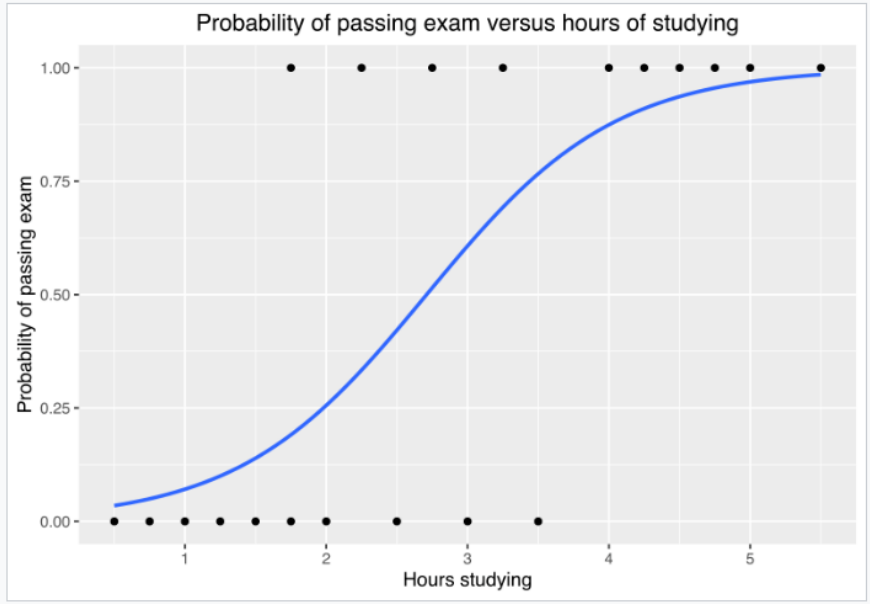


Modified example from Wikipedia page on Logistic Regression

Supervised ML: Logistic Regression

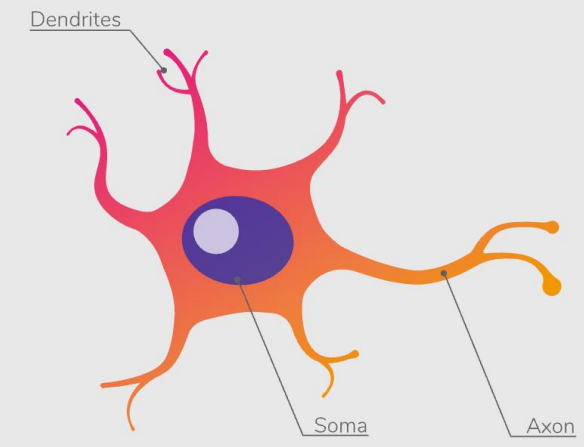
Hours studied	Result
0.3, 0.5, 0.7, 1.0, 1.2, 1.4, 1.7	Fail
2.3, 2.5, 2.7, 3.0, 3.4, 3.6, 3.9, 4.2, 4.5	Pass

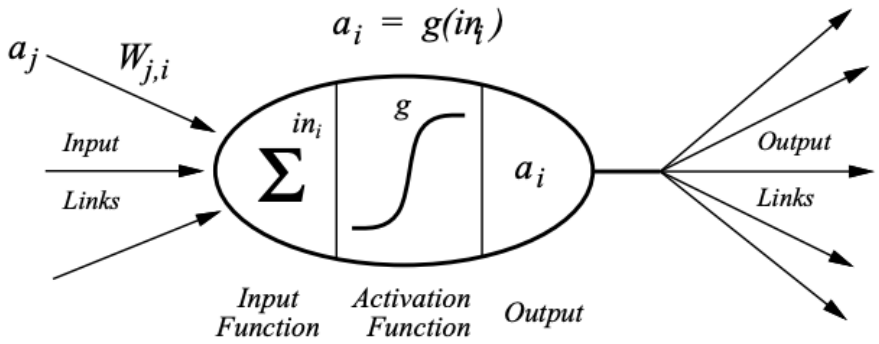


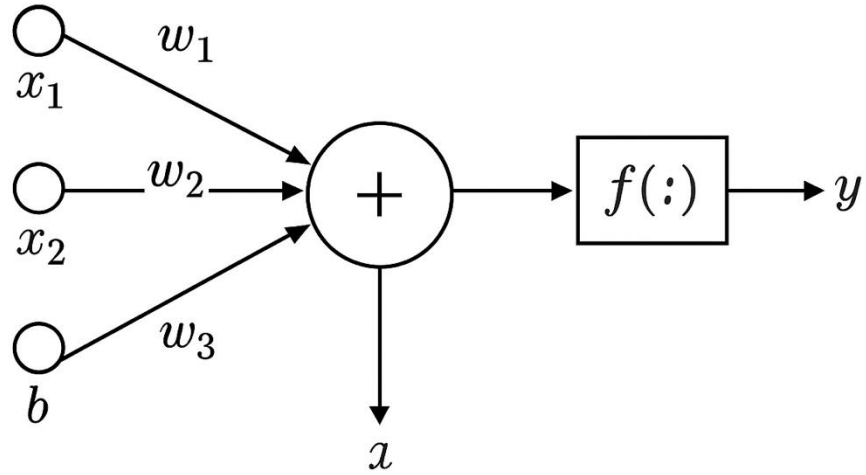


Graph of a logistic regression curve fitted to the (x_m, y_m) data. The curve shows the probability of passing an exam versus hours studying.

Modified example from Wikipedia page on Logistic Regression







$$a_i = g(\sum_j W_{j,i} a_j)$$

Inputs+ Bias + weights → hidden layers → predictions

Questions to consider for The Weight Adjustment Game:

- If learning for perceptrons means adjusting weights to minimize bias…
- **HOW do we get OPTIMAL weights for our model?**

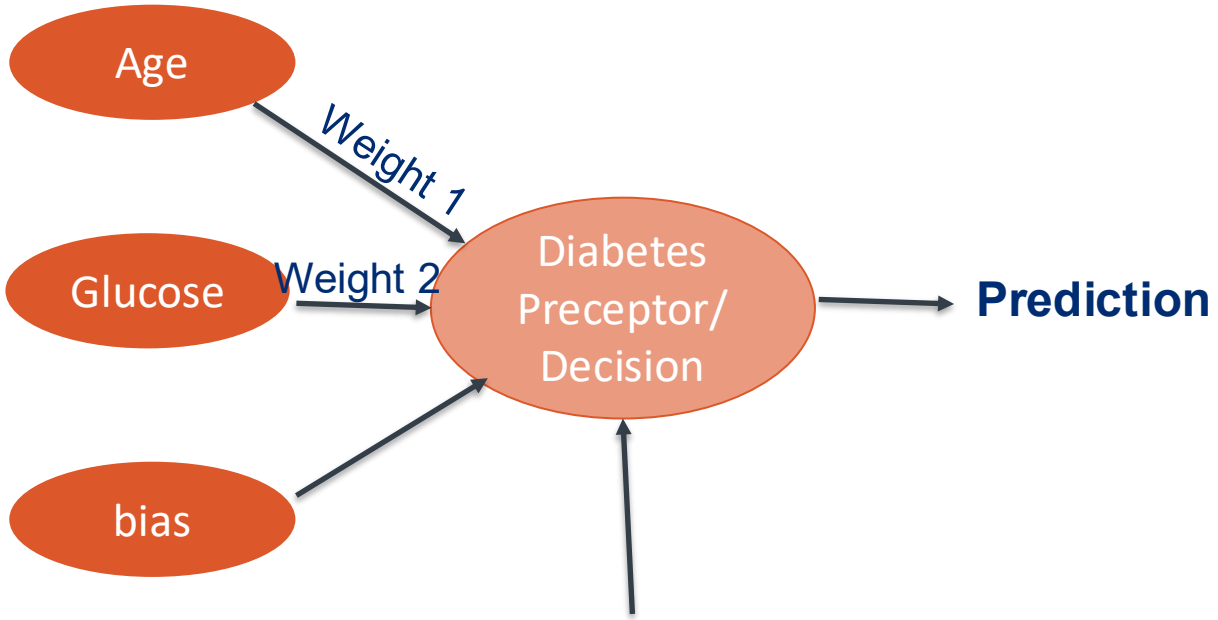
The Weight Adjustment Game

Age	Glucose	Has Diabetes (Yes = 1, No = 0)
40	130	1
25	100	0

The Weight Adjustment Game

- How difficult is it to determine weights?
- Bad weights = bad decisions

Age	Glucose	Has Diabetes (Yes = 1, No = 0)
40	130	1
25	100	0



Logistic: $\frac{1}{1+e^{-(w1*age+w2*Glucose+bias)}}$ >0.5 → diabetes

Example: w_1=0.1; w_2=0.02; bias=-5. Please try one case in the Google document (linked below).

Person 1: 0.01*40+0.02*130-5 = -2.0

$\frac{1}{1+e^{2.0}}$ = 0.119 → No diabetes predicted (incorrect)

Person 2: 0.01*25+0.02*100-5 = -2.75

$\frac{1}{1+e^{2.75}}$ = 0.060 → No diabetes predicted (correct)

https://docs.google.com/document/d/1hfhE7RphIAWiS8d-Kg5WGbxZIHLggDgq92w1vpG4Lus/edit?usp=sharing

Fair Treasure Splitting Game

1. Your group will receive one set of diamonds
2. You need to split the diamonds into 2 groups
3. Calculate the total value of each pile
4. Determine the **loss** (difference in values)
5. The team with the smallest loss wins

Think about:

- What was your strategy for dividing the diamonds?
- Was your split fair?
- If given more time, is there another approach that you might take?

Fair Treasure Splitting Game: de-brief

Thing to think about:

- What was your strategy for dividing the diamonds?
- Was your split fair?
- If given more time, is there another approach that you might take?

I will copy this into the chat

https://docs.google.com/presentation/d/1WURIjJi-HsY-gA5c86gcx1tat6yTmO_KxwZOx6VYYFA/edit?usp=sharing

This is where we finished on Day 1 – current summary

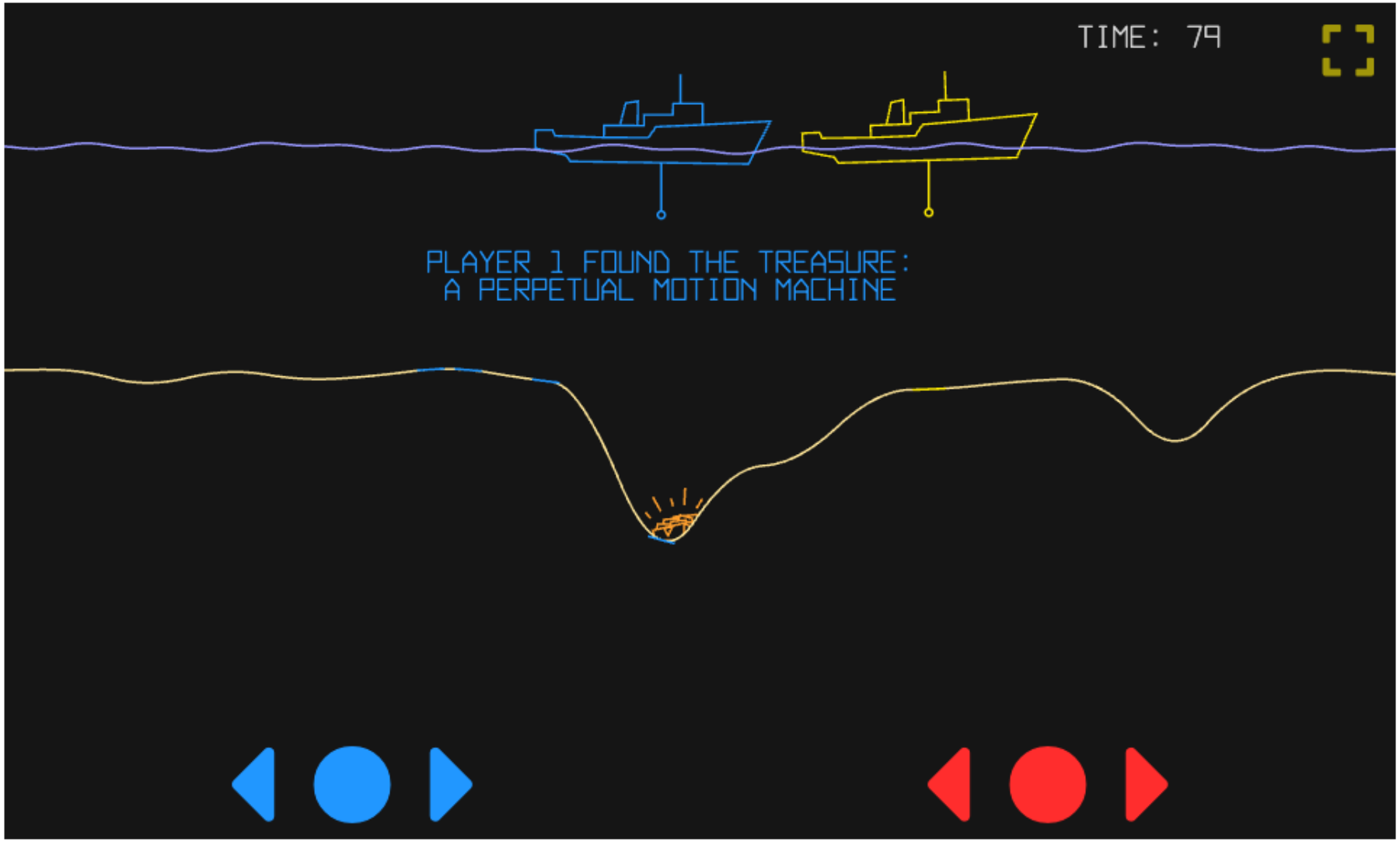
- Is ***emulation*** a sufficient substitute for "learning"?
  - Chinese Room thought experiment (also: what is learning?)

- We pick features (imperfectly) that allow us to model the world
  - Data needs to be in a format that can be utilized by computers
  - We make choices about data that compound over its lifecycle

- Regression is a model for **prediction** and **decisions**
  - Linear → OLS minimizes distance
  - Logistic → Threshold with a probability → plausibility

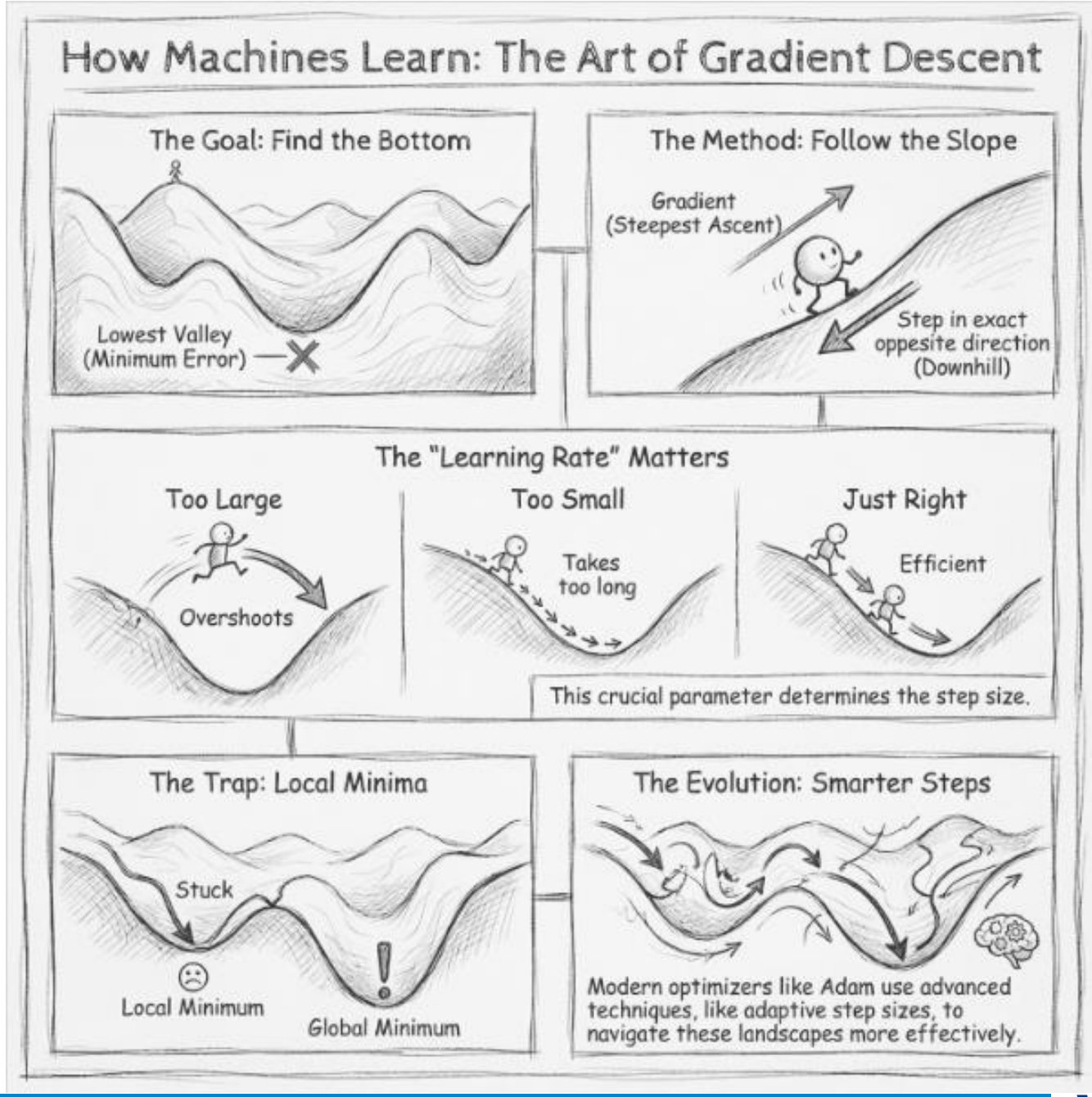
- **Loss** needs to be minimized in a model
  - How to split unequal gold as fairly as possible?

Gradient Descent



https://www.i-am.ai/gradient-descent.html

Gradient Descent



You are an Ai system. Your only job is to predict a number. Pick a number between 0-100.

You are an Ai system. Your only job is to predict a number. Pick a number between 0-100.

Now I will tell you the loss function.

Loss = |prediction -50|

What are people's choices?

You are an Ai system. Your only job is to predict a number. Pick a number between 0-100.

What if the loss is now calculated this way:

Loss = (prediction -50)^2

Would you change your original choice?

You are an Ai system. Your only job is to predict a number. Pick a number between 0-100.

What if the loss is now calculated this way:

Loss:

- **If prediction > 50 → penalty x 5**
- **if prediction <50 → normal**

Which number would you pick now and why?

You are an Ai system. Your only job is to predict a number. Pick a number between 0-100.

De-brief

- You didn’t become a different Ai system between rounds
- There was no new information given
- **The only change is what you were ‘punished’ for**

Questions to consider for the following simulation:

- Why are k-means clustering *unsupervised?*
- Can unsupervised algorithms uncover real patterns?
- How does k-means demonstrate how learning is done via algorithms?
- Are these clusters ‘real’?
- Parameter ‘knobs:
  - Does it matter how many k are chosen (k=2,k=3,etc)
  - Do different initial placements change the results?

Unsupervised ML: Classification

Visualizing K-Means Clustering

January 19, 2014

Suppose you plotted the screen width and height of all the devices accessing this website. You'd probably find that the points form three clumps: one clump with small dimensions, (smartphones), one with moderate dimensions, (tablets), and one with large dimensions, (laptops and desktops). Getting an algorithm to recognize these clumps of points without help is called *clustering*. To gain insight into how common clustering techniques work (and don't work), I've been making some visualizations that illustrate three fundamentally different approaches. This post, the first in this series of three, covers the k-means algorithm. To begin, click an initialization strategy below:

How to pick the initial centroids?

I'll Choose

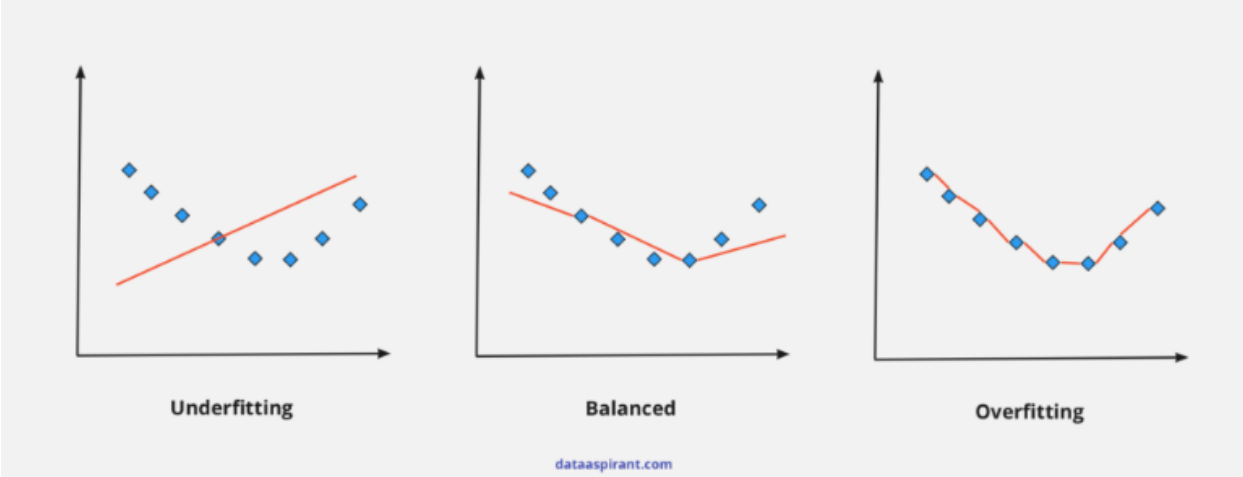
Randomly

Farthest Point

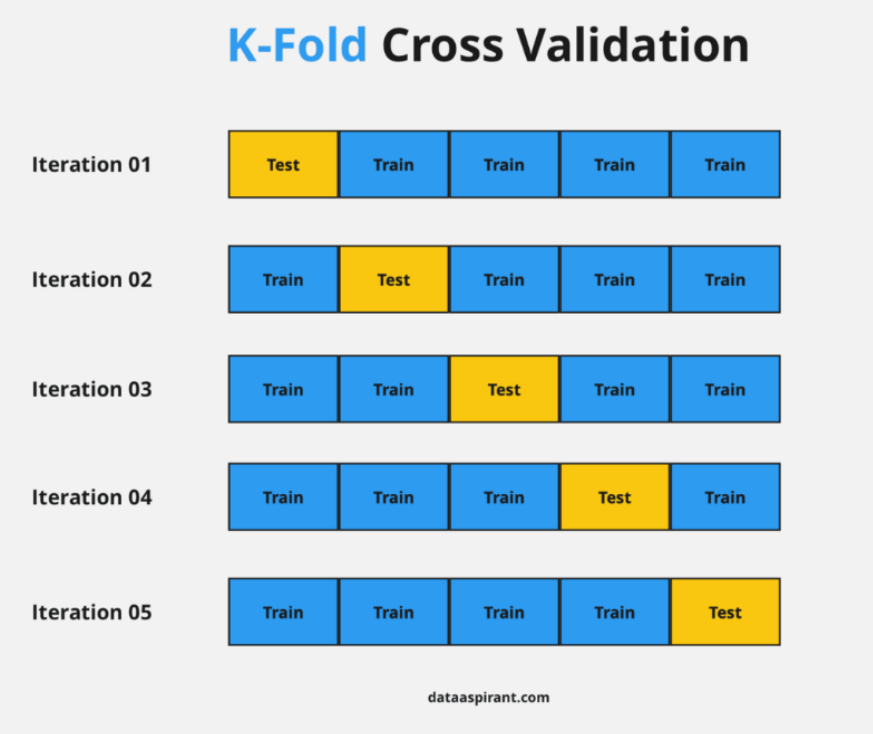
https://www.naftaliharris.com/blog/visualizing-k-means-clustering/

Training versus Testing

- Strategy: split the data and use some of it for **training** the NN to **make predictions** and **reserve** some of it **to test** the NN



- K-fold cross-validation



https://dataaspirant.com/cross-validation/

Hours Studied	Result
0.5	Fail
0.75	Fail
1.0	Fail
1.25	Fail
1.5	Fail
1.75	Fail
1.75	Pass
2.0	Fail
2.25	Pass
2.5	Fail
2.75	Pass
3.0	Fail
3.25	Pass
3.5	Fail

Hours Studied	Result
4.0	Pass
4.25	Pass
4.5	Pass
4.75	Pass
5.0	Pass
5.5	Pass

Training	Testing
1,2,3,4	5
1,2,3,5	4
1,2,4,5	3
1,3,4,5	2
2,3,4,5	1

K=5

Fold 1: 0.5, 2.25, 3.25, 4.75
Fold 2: 0.75, 1.75, 2.75, 4.5
Fold 3: 1.0, 2.5, 3.0, 4.25
Fold 4: 1.5, 1.75, 4.0, 5.5
Fold 5: 1.25, 2.0, 3.5, 5.0

5-k and 10-k folds are the most common

Synthesis of today's material

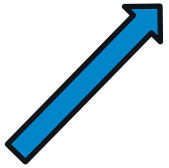
1. Ai isn't 'new'; it is mostly Machine Learning
2. Stacked logistic regressions approximates neural networks
3. "Learning" is minimizing loss; Gradient Descent is a convenient way to minimize loss.

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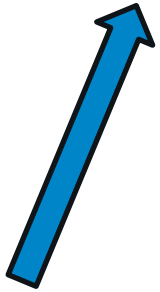
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4. Ai systems = Data x (Representation ⇒ **Objective/Loss**)x **Optimization**

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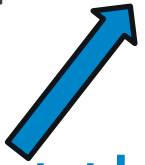
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How data is encoded



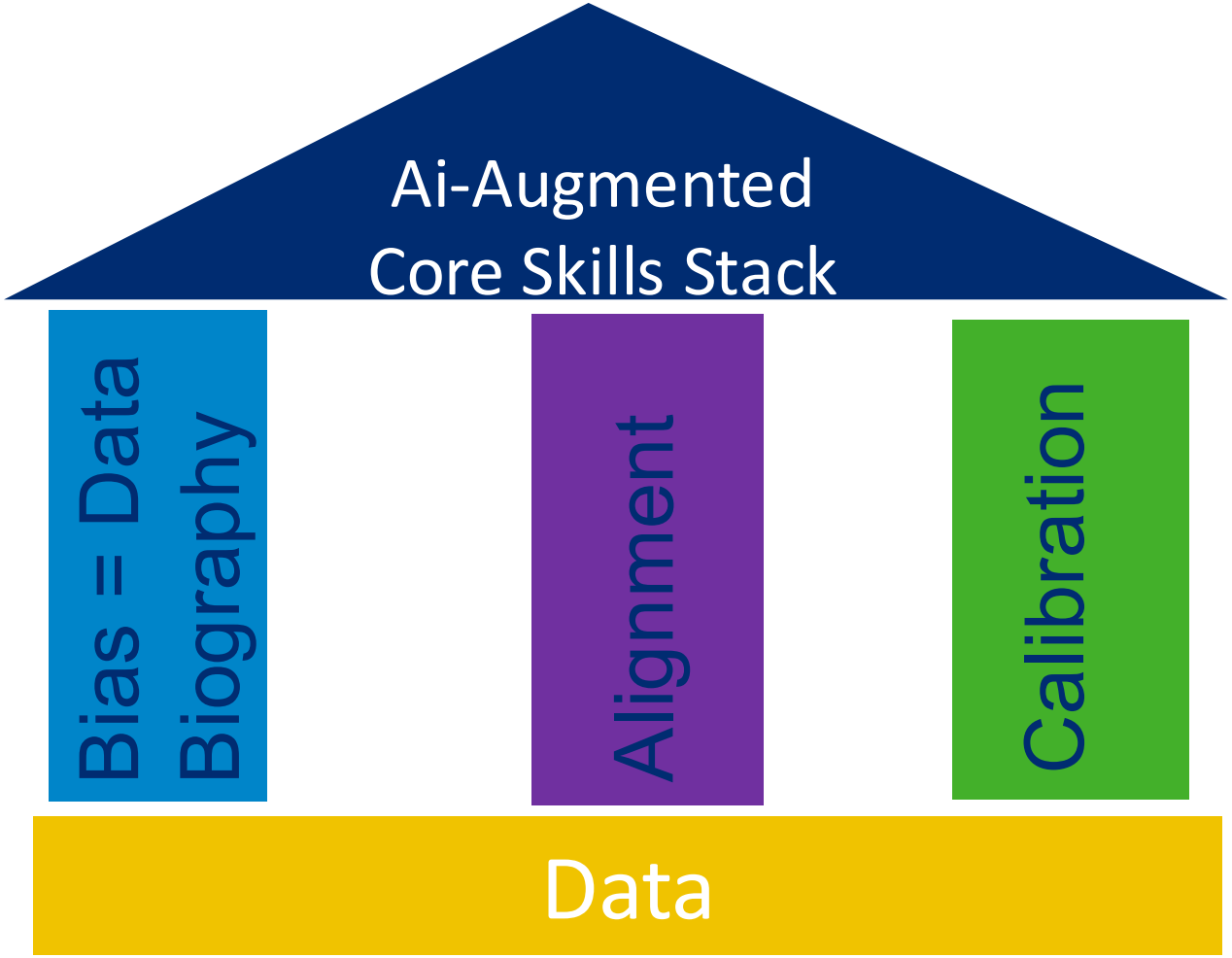
What we trying to achieve



How we get there

Synthesis of today's material

1. Ai isn't 'new'; it is mostly Machine Learning
2. Stacked logistic regressions approximates neural networks
3. Ai systems = Data x (Representation ⇒ Objective/Loss)x Optimization
4. Ai systems behave according to what they are rewarded for (demonstrated by their loss function) **not what we wish they did**



Ai = Data x (Representation ⟹ Objective/Loss) x Optimization

Exit Survey – 4 questions

I will post this in the chat:

https://forms.office.com/r/VHDshVvzrz